Bangladesh University of Engineering and Technology
Department of Computer Science & Engineering

Linear Algebra Question Bank

Linear Algebra & Matrix Theory

MATH 143 · MATH 243 · MATH 247 · MATH 259 · MATH 215

Year Coverage: 2010–2024

Topics: Vectors & Matrices, Linear Systems, Vector Spaces, Eigenvalues, Linear Transformations & more

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Linear Algebra & Matrix Theory

Vectors & Matrices

Properties, Special Matrices, Partitioning, Applications

S1 Vectors & Matrices

Q1. Determine whether the following statements are true or false, and justify your answer: (i) Every matrix has a unique echelon form. (ii) All leading 1's in a matrix in row echelon form must occur in different columns. (iii) If a linear system has more unknowns than equations, then it must have infinitely many solutions. (iv) For every square matrix A, it is true that $\mathrm{tr}(A^T) = \mathrm{tr}(A)$. (v) If B has a column of zeros, then so does AB if this product is defined. [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

Q2. A company produces Web design, software, and networking services. View the company as an open economy described by the table below, where input is in dollars needed for \$1.00 of output.

Provider	Web Design	Software	Networking
Web Design	\$0.40	\$0.20	\$0.45
Software	\$0.30	\$0.35	\$0.30
Networking	\$0.15	\$0.10	\$0.20

(i) Find the consumption matrix for the company. (ii) Suppose that the open sector demands \$5400 of Web design, \$2700 of software, and \$900 of networking. Use row reduction to find a production vector that meets this demand exactly. [MATH 143 | L-1/T-2 | 2023-24] **(15 marks)**

Q3. Define Markov Chain. Find the steady-state vector of the transition matrix $T = \begin{pmatrix} 0.5 & 0.4 & 0.6 \\ 0.2 & 0.2 & 0.3 \\ 0.3 & 0.4 & 0.1 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2023-24] **(15 marks)**

Q4. Let the vertex matrix of a directed graph be $A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$. (i) Draw a diagram of the directed graph. (ii) Find the number of 1-, 2-, and 3-step connections from vertex P_4 to vertex P_3. [MATH 143 | L-1/T-2 | 2023-24] **(13 marks)**

Q5. Decode the message 'GTNKGKDUSK' given that it is a Hill cipher with enciphering matrix $E = \begin{pmatrix} 5 & 6 \\ 2 & 3 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

Q6. Define regular Markov chain. Fill in the missing entries of the stochastic matrix $P = \begin{pmatrix} \frac{7}{10} & * & \frac{1}{5} \\ * & \frac{3}{10} & * \\ \frac{1}{10} & \frac{3}{5} & \frac{3}{10} \end{pmatrix}$ whose column sum is one, and find its steady-state vector. [MATH 143 | L-1/T-2 | 2022-23] **(15 marks)**

Q7. Let M be the following vertex matrix of a directed graph: $M = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$. (i) Draw a diagram of the directed graph. (ii) Use r-step connections to find the number of 1-, 2-, and 3-step connections from P_1 to P_2. Verify your answer by listing the various connections. [MATH 143 | L-1/T-2 | 2022-23] **(13 marks)**

Q8. Decode the message 'SAKNOXAOJX' given that it is a Hill cipher with enciphering matrix $\begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2022-23] **(10 marks)**

Q9. Use the enciphering matrix $\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$ to obtain the Hill cipher of the plaintext message 'DARK NIGHT'. [MATH 143 | L-1/T-2 | 2021-22] **(10 marks)**

Q10. Write down the statement of the Cayley-Hamilton theorem and use its idea to calculate A^n where $A = \begin{pmatrix} 1 & 3 \\ 4 & 7 \end{pmatrix}$ and n is any positive integer. [MATH 259 | 2020–21 (BUET)]

Q11. Prove that every square matrix A can be expressed as the sum of a symmetric and skew-symmetric matrix. [MATH 259 | L-2/T-1 | 2018-19] **(11 marks)**

↪ *Similar: MATH 259 | 2020-21; MATH 215 | 2018-19*

Q12. Find an upper triangular matrix that satisfies $A^3 = \begin{pmatrix} 1 & 30 \\ 0 & -8 \end{pmatrix}$, if possible. [MATH 259 | L-2/T-1 | 2018-19] **(11 marks)**

Q13. Prove that every square matrix can be expressed in one and only one way as the sum of a symmetric matrix and a skew-symmetric matrix. [MATH 215 | L-2/T-2 | 2018-19] **(10 marks)**

Q14. Define symmetric and skew-symmetric matrices. Show that the inverse of a non-singular matrix is unique. [MATH 215 | L-2/T-2 | 2017-18] **(10 marks)**

Q15. Find the adjoint of the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. Hence find $P(A)$ when $P(x) = x + 2x^{-1} - 4$. [MATH 215 | L-2/T-2 | 2017-18] **(15 marks)**

Q16. Describe briefly ten different fields of applications of Linear Algebra. [MATH 215 | L-2/T-2 | 2017-18] **(10 marks)**

Q17. Discuss how the rank of A varies with t: $A = \begin{pmatrix} t & 3 & -1 \\ 3 & 6 & -2 \\ -1 & -3 & t \end{pmatrix}$. [MATH 259 | 2017–18 (BUET)]

Q18. State Cayley-Hamilton theorem. Find A^{-1} using this theorem, where $A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$. [MATH 215 | 2017–18]

Q19. What do you mean by an elementary matrix? Express the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 5 & 5 & 1 \end{pmatrix}$ as a product of elementary matrices. [MATH 215 | 2017–18]

Q20. Find an LU factorization of $A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 8 \\ 2 & 1 & 5 \end{pmatrix}$. [MATH 215 | 2017–18]

Q21. Define Hermitian and Nilpotent matrices with examples. [MATH 215 | L-2/T-2 | 2016-17] **(18 marks)**

Q22. Define idempotent matrix and nilpotent matrix with examples. [MATH 259 | L-2/T-1 | 2015-16] **(20 part marks)**

Q23. Find AB by conformal partitioning of A and B, given $A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 \\ 2 & 0 & -4 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 & 2 & 2 \\ 1 & 3 & 0 & 0 & 1 & 0 \\ -3 & -1 & 2 & 1 & 0 & -1 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q24. For the matrix $A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$, using Cayley-Hamilton theorem find the value of A^n. [MATH 259 | 2015–16 (BUET)]

Q25. Define orthogonal and idempotent matrices with examples. Prove that the only matrices which commute with *every* $n \times n$ matrix are the $n \times n$ scalar matrices. Compute AB by the method of conformal partitioning, when $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 3 & 1 & 2 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2014-15] **(18 marks)**

Q26. Show that A is non-singular and find the left and right quotients of B by A, where $B = \begin{pmatrix} 2 & 1 & 5 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 6 & 7 & 8 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2014-15] **(17 marks)**

Q27. Find the volume of the parallelepiped whose edges are represented by $\mathbf{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\mathbf{b} = \hat{i} + 2\hat{j} - \hat{k}$, and $\mathbf{c} = 3\hat{i} - \hat{j} + 2\hat{k}$. [MATH 243 | L-2/T-2 | 2014-15] **(12 marks)**

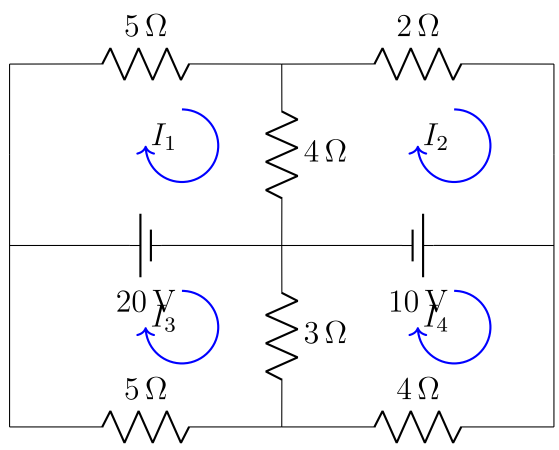
Q28. If A and B are symmetric matrices, prove that AB is symmetric if and only if A and B commute. [MATH 259 | L-2/T-1 | 2013-14] **(10 marks)**

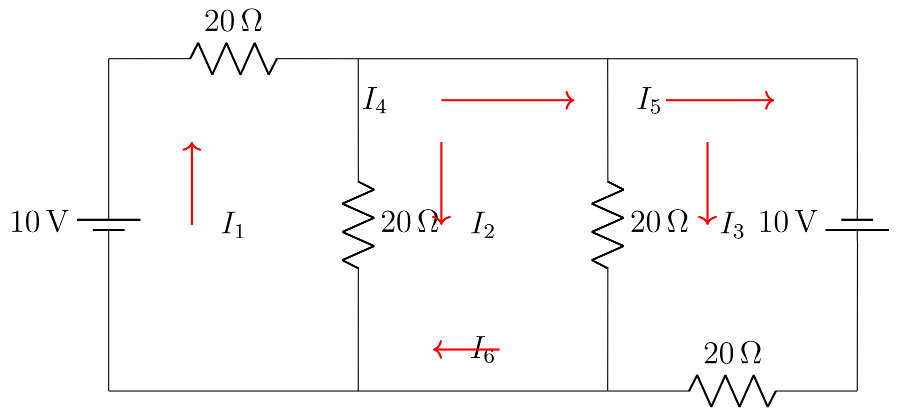
Q29. If A and B are non-singular square matrices, show that $\operatorname{adj}(AB) = \operatorname{adj}B \cdot \operatorname{adj}A$. [MATH 259 | L-2/T-1 | 2013-14] **(15 marks)**

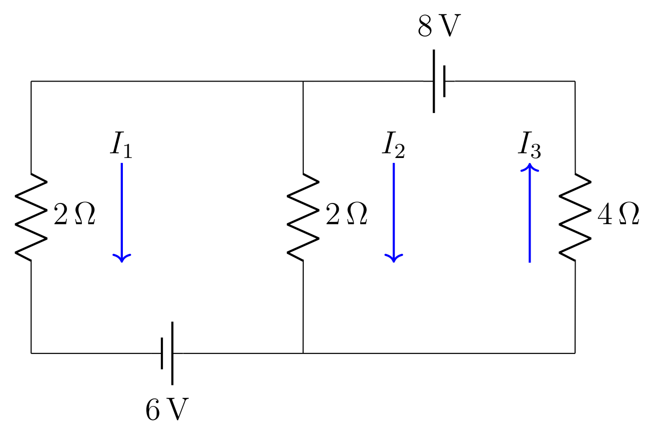
Q30. Find the adjoint of the matrix $A = \begin{pmatrix} 9 & 2 & 4 \\ 6 & 7 & 3 \\ 1 & 5 & 8 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2013-14] **(10 marks)**

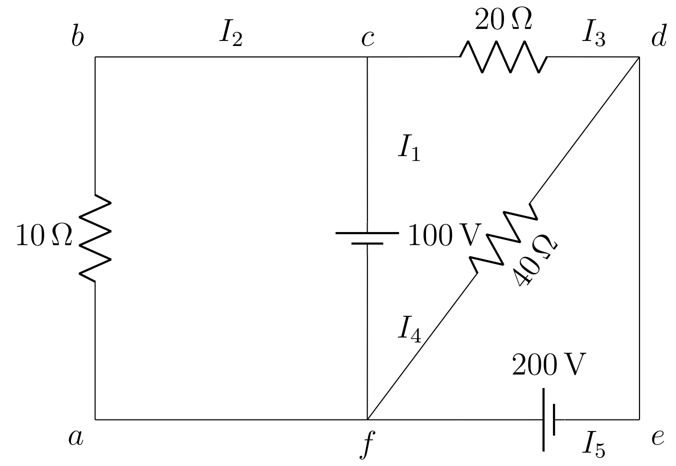
Q31. Prove that the volume of the parallelepiped whose adjacent edges are $\mathbf{a} \times \mathbf{b}$, $\mathbf{b} \times \mathbf{c}$, and $\mathbf{c} \times \mathbf{a}$ is equal to the square of the volume of the parallelepiped whose adjacent edges are $\mathbf{a}, \mathbf{b}, \mathbf{c}$. [MATH 243 | L-2/T-2 | 2013-14] **(12.5 marks)**

Q32. Find the unknown currents in the given circuits using Kirchhoff's Laws. [MATH 259 | L-2/T-1 | 2013-14, 2014-15, 2017-18, 2018-19, 2020-21] **(15 marks)**









Q33. Consider the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $P(x) = x + 2x^{-1} - 4$. Find $P(A)$. [MATH 243 | L-2/T-2 | 2011-12] **(15 marks)**

Q34. Prove that $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})^2$. [MATH 243 | L-2/T-2 | 2011-12] **(10 marks)**

Q35. Show that the four points whose position vectors are $3\hat{i} - 3\hat{j} + 4\hat{k}$, $6\hat{i} + 3\hat{j} + \hat{k}$, $5\hat{i} + 7\hat{j} + 3\hat{k}$ and $2\hat{i} + 2\hat{j} + 6\hat{k}$ are coplanar. [MATH 243 | L-2/T-2 | 2011-12] **(10 marks)**

Q36. Write down the geometrical interpretation of dot product and cross product of vectors. For the vectors **A**, **B** and **C** prove that $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$. [MATH 243 | L-2/T-2 | 2010-11] **(23 marks)**

Q37. Find the minimal and characteristic equations of the matrix $A = \begin{pmatrix} 5 & 4 & -1 \\ 4 & 5 & -1 \\ -4 & -4 & 3 \end{pmatrix}$. Is the matrix A derogatory or not? If yes, express the characteristic polynomial as the product of the minimal polynomial and one of its monic factors. [MATH 259 | Classic]

Q38. Reduce the matrix $A = \begin{pmatrix} 2 & 7 & 3 & 5 \\ 1 & 2 & 3 & 4 \\ 3 & 8 & 1 & -2 \\ 4 & 13 & 1 & -1 \end{pmatrix}$ to echelon form then to its canonical form. Write down the rank and nullity. Also verify the Dimension theorem. [MATH 259 | Classic]

BUET · Linear Algebra & Matrix Theory

Linear Algebra & Matrix Theory

Solving Linear Systems

Row Reduction, LU Factorization, Elementary Matrices, Rank

S2 Solving Linear Systems

Q39. By examining the determinant of the coefficient matrix, find the condition for which the following system has a nontrivial solution: $x + y + \alpha z = 0, \quad x + y + \beta z = 0, \quad \alpha x + \beta y + z = 0.$ [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

Q40. Define the rank of a matrix. Find the values of r and s for which $\begin{pmatrix} 1 & 0 & 0 \\ 0 & r-2 & 2 \\ 0 & s-1 & r+2 \\ 0 & 0 & 3 \end{pmatrix}$ has rank 1 and rank 2. Justify your answer. [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

↪ *Similar: MATH 259 | 2020-21*

Q41. Find an LU decomposition of the matrix $A = \begin{pmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2023-24] **(13 marks)**

↪ *Similar: MATH 143 | 2021-22*

Q42. Solve the following system of linear equations after reducing the augmented matrix to its reduced row-echelon form: $x-y+2z-w = -1$, $2x+y-2z-2w = -2$, $-x+2y-4z+w = -1$, $3x-3w = -3$. [MATH 143 | L-1/T-2 | 2022-23] **(20 marks)**

Q43. Determine the values of k so that the following system has: (i) no solution, (ii) more than one solution, (iii) infinitely many solutions: $x + y - z = 1, \quad 2x + 3y + kz = 3, \quad x + ky + 3z = 2.$ [MATH 143 | L-1/T-2 | 2022-23] **(15 marks)**

Q44. Find an LU decomposition of the matrix $A = \begin{pmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2022-23] **(13 marks)**

Q45. Define least squares solution, the least squares error vector, and the least squares error. Find the least squares solutions, the least squares error vector, and the least squares error of the linear system: $3x + 2y - z = 2, \quad x - 4y + 3z = -2, \quad x + 10y - 7z = 1$. Verify that the least squares error vector is orthogonal to the column space of the system. [MATH 143 | L-1/T-2 | 2022-23] **(15 marks)**

Q46. Solve the system of homogeneous equations by reducing the coefficient matrix to its reduced row echelon form: $2x + 2y + 4z = 0$, $-2x - 2y - z = 0$, $2w + 3x + y + z = 0$, $-2w + x + 3y - 2z = 0$. [MATH 259 | L-2/T-1 | 2021-22] **(18 marks)**

Q47. Determine the values of a for which the system has no solutions, unique solution, or infinitely many solutions. $x + 2y + z = 2$, $2x - 2y + 3z = 1, \quad x + 2y - (a^2 - 3)z = a$. [MATH 259 | L-2/T-1 | 2021-22] **(17 marks)**

Q48. Reduce the matrix $A = \begin{pmatrix} 2 & 7 & 3 & 5 \\ 1 & 2 & 3 & 4 \\ 3 & 8 & 1 & -2 \\ 4 & 13 & 1 & -1 \end{pmatrix}$ to echelon form, then to its canonical form. Write down the rank and nullity. Also verify the Dimension Theorem. [MATH 259 | L-2/T-1 | 2021-22] **(18 marks)**

Q49. Solve the system of homogeneous equations by reducing the coefficient matrix to its canonical form (Gauss-Jordan elimination): $x + 2y + 3z + t = 0$, $2x + 4y + 7z + 4t = 0$, $3x + 6y + 10z + 5t = 0$. Write down a basis and dimension of the solution space. [MATH 143 | L-1/T-2 | 2021-22] **(18 marks)**

Q50. Determine the values of a for which the system $x + 2y - 3z = 4$, $3x - y + 5z = 2$, $4x + y + (a^2 - 14)z = a + 2$ has no solution, unique solution, or infinitely many solutions. Justify your answer. [MATH 143 | L-1/T-2 | 2021-22] **(17 marks)**

Q51. Write down the properties of triangular matrices with examples. Find an LU-decomposition of the matrix $A = \begin{pmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2021-22] **(17 marks)**

Q52. What is an elementary matrix? With the help of elementary matrices find the LU decomposition of the matrix $\begin{pmatrix} 2 & 1 & -1 \\ -2 & -1 & 2 \\ 2 & 1 & 0 \end{pmatrix}$. Also make necessary amendment to the statement: "Every square matrix has an LU decomposition". [MATH 259 | L-2/T-1 | 2020-21] **(17 marks)**

Q53. Which types of solution should one expect from a system of linear equations when the determinant of the coefficient matrix is zero? Write your answer with relevant examples. [MATH 259 | L-2/T-1 | 2020-21] **(5 marks)**

Q54. Solve the following system of linear equations by finding the inverse of the coefficient matrix using elementary row operations: $2x - y + 3z + 4w = 9, \quad x - 2z + 7w = 11, \quad 3x - 3y + z + 5w = 8, \quad 2x + y + 4z + 4w = 10.$ [MATH 259 | L-2/T-1 | 2020-21] **(15 marks)**

Q55. For the matrix $B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$, find non-singular matrices P and Q such that PBQ takes the normal form. Hence determine the rank of B. [MATH 259 | L-2/T-1 | 2020-21] **(15 marks)**

↪ *Similar: MATH 259 | 2021-22*

Q56. Solve the following system of linear equations (homogeneous): $x_1 + 3x_2 + x_4 = 0$, $x_1 + 4x_2 + 2x_3 = 0$, $-2x_2 - 2x_3 - x_4 = 0$, $2x_1 - 4x_2 + x_3 + x_4 = 0$, $x_1 - 2x_2 - x_3 + x_4 = 0$. [MATH 247 | L-2/T-2 | 2020-21] **(20 marks)**

Q57. Solve the following linear system by Gauss-Jordan elimination method: $x_1 + 3x_2 + x_4 = 0$, $x_1 + 4x_2 + 2x_3 = 0$, $-2x_2 - 2x_3 - x_4 = 0$, $2x_1 - 4x_2 + x_3 + x_4 = 0$, $x_1 - 2x_2 - x_3 + x_4 = 0$. [MATH 247 | 2020–21]

Q58. Solve the system of linear equations by Gauss-Jordan elimination: $x+2y-z=4$, $x-y=3$, $x+3y-2z=7$, $2u+4v+w+7x=7$. [MATH 247 | L-2/T-2 | 2019-20] **(20 marks)**

Q59. Determine the values of a for which the system has no solutions, exactly one solution, or infinitely many solutions. $x+y+7z=-7$, $2x+3y+17z=-16$, $x+2y+(a^2+1)z=3a$. [MATH 259 | L-2/T-1 | 2018-19] **(12 marks)**

Q60. Use the inversion algorithm to find the inverse of $A=\begin{pmatrix}2&-4&0&0\\1&2&12&0\\0&0&2&0\\0&-1&-4&-5\end{pmatrix}$. [MATH 259 | L-2/T-1 | 2018-19] **(12 marks)**

Q61. Solve the following system of equations by reducing the augmented matrix to its canonical form: $x-2y-3z=-4$, $2x+3y+4z=8$, $4x-2y-2z=2$. [MATH 247 | L-2/T-2 | 2018-19] **(20 marks)**

Q62. Find the rank and a basis of the row space of $A=\begin{pmatrix}2&4&-2&1&1\\2&5&4&-2&2\\4&3&1&1&2\\2&-4&2&-1&1\\0&1&4&2&-1\end{pmatrix}$. [MATH 215 | L-2/T-2 | 2017-18] **(15 marks)**

Q63. Find an LU factorization of $A=\begin{pmatrix}1&1&0\\4&6&1\\-2&2&0\end{pmatrix}$. [MATH 215 | L-2/T-2 | 2018-19] **(10 marks)**

Q64. Write the LU decomposition of the matrix $A=\begin{pmatrix}1&-3&0\\0&1&3\\2&-10&2\end{pmatrix}$. [MATH 259 | L-2/T-1 | 2017-18] **(10 marks)**

Q65. Define elementary matrix. Express the matrix $A=\begin{pmatrix}2&4&7&3\\5&-6&8&9\\7&4&3&8\end{pmatrix}$ in the form $B=E_iE_{i-1}\cdots E_2E_1A$, where B is the echelon form. [MATH 259 | L-2/T-1 | 2017-18] **(10 marks)**

Q66. Prove that if A is an invertible matrix, then the system of linear equations $Ax=b$ has a unique solution given by $x=A^{-1}b$. [MATH 259 | L-2/T-1 | 2017-18] **(7 marks)**

Q67. If $A=\begin{pmatrix}1&-1&0&2\\0&1&1&-1\\2&1&2&1\\3&2&1&6\end{pmatrix}$, find two non-singular matrices P and Q such that $PAQ=I$. Hence find A^{-1} in terms of P and Q. [MATH 259 | L-2/T-1 | 2017-18] **(20 marks)**

Q68. Solve the following linear system by Gaussian elimination: $3x_1+2x_2-x_3=-15$, $5x_1+3x_2+2x_3=0$, $3x_1+x_2+3x_3=11$, $-6x_1-4x_2+2x_3=30$. [MATH 247 | L-2/T-2 | 2017-18] **(20 marks)**

Q69. Solve the following system of linear equations (if possible): $-x+2y-4z+w=1$, $3x-3w=-3$. [MATH 247 | L-2/T-2 | 2017-18] **(10 marks)**

Q70. Solve the following system of linear equations: $2u+v-2w-2x=-2$, $-u+2v-4w+x=1$, $3u-3x=-3$. [MATH 247 | L-2/T-2 | 2017-18] **(10 marks)**

Q71. Solve the following system of linear equations (if possible) by reducing the augmented matrix into its reduced row echelon form: $-x+2y-4z+w=1$, $3x-3w=-3$. [MATH 215 | L-2/T-2 | 2018-19] **(10 marks)**

Q72. State Cayley-Hamilton theorem. Verify this theorem for the matrix $A=\begin{pmatrix}1&0&3\\2&1&-1\\1&-1&1\end{pmatrix}$ and hence express A^5 as a linear combination of A^2, A and I. [MATH 259 | L-2/T-1 | 2017-18] **(15 marks)**

Q73. Define elementary matrix. Find a sequence of elementary matrices that can be used to write the matrix $A=\begin{pmatrix}0&1&3\\1&-3&0\\2&-6&2\\0&1&4\end{pmatrix}$ in row-echelon form. [MATH 259 | L-2/T-1 | 2016-17] **(11 marks)**

Q74. Factorize the matrix $A=\begin{pmatrix}1&2&3\\2&5&3\\1&0&8\end{pmatrix}$ into LU. [MATH 259 | L-2/T-1 | 2016-17] **(14 marks)**

Q75. Check the consistency of the following system of linear non-homogeneous equations and find the solution, if it exists. $x_1+x_2+2x_3=9$; $2x_1+4x_2-3x_3=1$; $3x_1+6x_2-5x_3=0$. [MATH 259 | L-2/T-1 | 2016-17] **(10 marks)**

Q76. Solve the system of linear equations by Gaussian elimination: $x-y+z-r+s=1$, $2x-y+3z+4s=2$, $3x-2y+2z+r+s=1$, $x+z+2r+s=0$. [MATH 247 | L-2/T-2 | 2016-17] **(20 marks)**

Q77. If $A=\begin{pmatrix}3&-3&4\\2&-3&4\\0&-1&1\end{pmatrix}$, find two non-singular matrices P and Q such that $PAQ=I$. Hence find A^{-1}. [MATH 259 | L-2/T-1 | 2016-17] **(15 marks)**

Q78. For the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ 3 & 1 & 0 \end{pmatrix}$, verify that $A(\text{adj } A) = |A|I_3$. Find the inverse of A as well. [MATH 215 | L-2/T-2 | 2016-17] **(15 marks)**

Q79. Factorize the matrix $A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ into elementary matrices. [MATH 215 | L-2/T-2 | 2016-17] **(10 marks)**

Q80. Prove that a square matrix A is invertible if it can be written as a product of elementary matrices. Hence express $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{pmatrix}$ as a product of elementary matrices. [MATH 259 | L-2/T-1 | 2015-16] **(20 marks)**

Q81. Solve the linear system using LU-factorization: $2x_1 + 4x_2 - 2x_3 = 4, \quad 6x_1 + 3x_3 = 15, \quad 4x_1 + 2x_2 + 4x_3 = 6.$ [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q82. If $A = \begin{pmatrix} 2 & 4 & 6 \\ 4 & 9 & 12 \\ 0 & 10 & 1 \end{pmatrix}$, find two non-singular matrices P and Q such that $PAQ = I$. Hence find A^{-1}. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q83. Determine the value of k such that the following system has: (i) a unique solution, (ii) no solution, (iii) more than one solution: $kx + y + z = 1, \quad x + ky + z = 1, \quad x + y + kz = 1.$ [MATH 243 | L-2/T-2 | 2015-16] **(15 marks)**

Q84. Obtain the canonical matrix C row equivalent to $A = \begin{pmatrix} 2 & 4 & -3 & 6 & 4 \\ 1 & 2 & -2 & 3 & 1 \\ 1 & 3 & -2 & 3 & 0 \\ 1 & 1 & -1 & 4 & 6 \end{pmatrix}$. [MATH 215 | L-2/T-2 | 2015-16] **(15 marks)**

Q85. Show that elementary transformations do not alter the rank of a matrix. Solve the system of equations:

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 4 \\ 2x_1 - x_2 - x_3 + 3x_4 &= 6 \\ 3x_1 + 4x_2 - 5x_3 + 6x_4 &= -11 \\ 7x_1 - 5x_2 + 7x_3 + x_4 &= 46 \end{aligned}$$

[MATH 259 | L-2/T-1 | 2014-15] **(18 marks)**

Q86. Define canonical form of a matrix. Reduce the matrix $A = \begin{pmatrix} 1 & -2 & 1 & 3 \\ 4 & -1 & 5 & 8 \\ 2 & 3 & 3 & 2 \end{pmatrix}$ into canonical form. Also find its rank. [MATH 259 | L-2/T-1 | 2014-15] **(17 marks)**

↪ *Similar: MATH 259 | 2018-19*

Q87. Determine the relationships among the constants p, q, and r under which the following system has a unique solution and infinitely many solutions (if any): $x + 2y - 3z = p, \quad 3x - y + 2z = q, \quad x - 5y + 8z = r.$ [MATH 243 | L-2/T-2 | 2014-15] **(15 marks)**

Q88. If A and B are non-singular square matrices, prove that: (i) $(AB)^{-1} = B^{-1}A^{-1}$ and (ii) $(A^{-1})^T = (A^T)^{-1}$. [MATH 259 | L-2/T-1 | 2014-15] **(10 marks)**

Q89. Find the inverse of the matrix $\begin{pmatrix} 1 & 4 & 3 & 1 \\ 0 & 5 & 4 & 3 \\ 2 & 5 & 3 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2013-14] **(20 marks)**

Q90. Find an LU or PLU factorization of the matrix $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 8 \\ 2 & 1 & 5 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2013-14] **(15 marks)**

Q91. Solve the following system of linear equations by reducing the augmented matrix to its reduced row-echelon form: $x + y + 2z = 9, \quad 2x + 4y - 3z = -1, \quad 3x + 6y - 5z = -3$. Write the system in matrix form $AX = B$, find A^{-1} by elementary row operations, and find X using A^{-1}. [MATH 243 | L-2/T-2 | 2013-14] **(15 marks)**

Q92. Find the adjoint of the matrix $A = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{pmatrix}$ and hence find A^{-1}. Also verify that $AA^{-1} = I_3$. [MATH 243 | L-2/T-2 | 2013-14] **(15 marks)**

Q93. Use elementary row transformations to find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 5 \\ 4 & 5 & 5 & 7 \end{pmatrix}$. [MATH 243 | L-2/T-2 | 2012-13] **(23.33 marks)**

Q94. Solve the following system of linear equations: $x_1 + 2x_2 - x_3 - 2 = 0, \quad 3x_1 + x_2 + 2x_3 - 11 = 0, \quad 4x_1 + 4x_2 - 3x_3 - 3 = 0, \quad 2x_1 - x_2 + 3x_3 - 9 = 0.$ [MATH 243 | L-2/T-2 | 2012-13] **(15 marks)**

Q95. Reduce the matrix $A = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 3 & 1 & -2 & -1 \\ 4 & -3 & 1 & 1 \\ 3 & -2 & 0 & -1 \end{pmatrix}$ to canonical form and hence find the rank. [MATH 243 | L-2/T-2 | 2012-13] **(15 marks)**

Q96. Solve the following system by reducing the augmented matrix to its canonical form: $3x + 5y - 7z = 13$, $4x + y - 12z = 6$, $2x + 9y - 3z = 20$. [MATH 243 | L-2/T-2 | 2011-12] **(15 marks)**

Q97. Find non-singular matrices P and Q such that PAQ is in normal form, where $A = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 3 & 1 & -2 & -1 \\ 4 & -3 & 1 & 1 \end{pmatrix}$. [MATH 243 | L-2/T-2 | 2011-12] **(20 marks)**

Q98. Find the inverse of the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix}$ by adjoint method. Then show that $A(\text{adj}A) = |A|I_3$. [MATH 243 | L-2/T-2 | 2010-11] **(20 marks)**

Q99. Solve the following system of equations by reducing the augmented matrix to its canonical form: $x+y+z+w = 4$, $2x-y-z+3w = 6$, $3x + 4y - 5z + 6w = -11$, $7x - 5y + 7z + w = 46$. [MATH 243 | L-2/T-2 | 2010-11] **(26.67 marks)**

Q100. Find non-singular matrices P and Q such that PAQ is in the normal form, where $A = \begin{pmatrix} 3 & 2 & -1 & 5 \\ 5 & 1 & 4 & -2 \\ 1 & -4 & 11 & -19 \end{pmatrix}$. [MATH 259 | Classic]

Q101. Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$. Hence find A^{-1}. [MATH 259 | L-2/T-1 | Classic] **(10 marks)**

Q102. Find the inverse of the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{pmatrix}$ by adjoint method. Then show that $A(\text{adj}\, A) = |A|I_3$. [MATH 243 | L-2/T-2 | Classic] **(15 marks)**

Q103. For the matrix $A = \begin{pmatrix} 1 & 3 & -1 & 3 \\ 3 & 4 & -3 & 4 \\ 1 & 3 & 1 & 2 \end{pmatrix}$, find non-singular matrices P and Q such that PAQ is in normal form. [MATH 259 | L-2/T-1 | Classic] **(15 marks)**

BUET · Linear Algebra & Matrix Theory

Linear Algebra & Matrix Theory

Vector Spaces

Subspaces, Basis, Dimension, Span, Null Space, Rank

S3 Vector Spaces

Q104. Determine whether the subset $W = \{$all vectors of the form (a, b, c, d), where $a = b = c = d\}$ is a subspace of $\mathbb{R}^4$. If so, find the dimension of W. [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

↪ *Similar: MATH 259 | 2020-21*, MATH 215 | 2016-17

Q105. Define basis and dimension of a vector space. [MATH 143 | L-1/T-2 | 2022-23] **(4 marks)**

Q106. Determine whether the following subsets are subspaces of $\mathbb{R}^4$. Find a basis and dimension: (i) all vectors (a, b, c, d) where $d = a+b$ and $c = a - b$; (ii) all vectors (a, b, c, d) where $d = 2a + 7c$ and $3c = 2a - 5b$. [MATH 143 | L-1/T-2 | 2022-23] **(16 marks)**

↪ *Similar: MATH 259 | 2016-17, 2018-19; MATH 247 | 2019-20; MATH 215 | 2016-17*

Q107. Find a subset of vectors $u_1 = (1, -1, 5, 2)$, $u_2 = (-2, 3, 1, 0)$, $u_3 = (4, -5, 9, 4)$, $u_4 = (0, 4, 2, -3)$, $u_5 = (-7, 18, 2, -8)$ forming a basis for their span; express non-basis vectors as linear combinations of the basis. [MATH 143 | L-1/T-2 | 2022-23] **(15 marks)**

Q108. Test whether the following sets are subspaces of $\mathbb{R}^3$: (i) All vectors of the form $(a, 1, 1)$. (ii) All vectors of the form (a, b, c), where $b = a + c$. [MATH 143 | L-1/T-2 | 2021-22] **(18 marks)**

Q109. (i) Find a subset of vectors $v_1 = (1, -2, 0, 3)$, $v_2 = (2, -5, -3, 6)$, $v_3 = (0, 1, 3, 0)$, $v_4 = (2, -1, 4, -7)$, $v_5 = (5, -8, 1, 2)$ that forms a basis for the space spanned by these vectors. (ii) Express each non-basis vector as a linear combination of the basis vectors. [MATH 143 | L-1/T-2 | 2021-22] **(17 marks)**

Q110. Find a subset of vectors $u_1 = (1, -2, 0, 3)$, $u_2 = (2, -5, -3, 6)$, $u_3 = (-4, 11, 9, -12)$, $u_4 = (2, -1, 4, -7)$, $u_5 = (5, -8, 1, 2)$ forming a basis for their span; express non-basis vectors as linear combinations. [MATH 143 | L-1/T-2 | 2021-22] **(15 marks)**

Q111. Let $W = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} : a + b = 3c,\ b + c = 4d \right\}$. Is W a subspace of $\mathbb{R}^4$? If so, find a basis and dimension. [MATH 259 | L-2/T-1 | 2021-22] [MATH 215 | L-2/T-2 | 2017-18]**(10 marks)**

Q112. Find a 3×3 matrix whose null space is (i) a point, (ii) a line, (iii) a plane. [MATH 259 | L-2/T-1 | 2020-21] **(15 marks)**

Q113. Find a subset of the vectors $v_1 = (-1, 3, 2, 4)$, $v_2 = (2, -7, -5, -9)$, $v_3 = (0, 2, 2, 2)$, $v_4 = (4, 0, 4, -4)$, $v_5 = (5, 1, 6, -4)$, $v_6 = (-3, 4, 1, 7)$ that forms a basis for the space spanned by these vectors. Express each non-basis vector as a linear combination of the basis vectors, and find their coordinate vectors. [MATH 259 | L-2/T-1 | 2020-21] **(20 marks)**

Q114. Find the canonical form, bases for row space and column space, the rank, and the nullity of the matrix $A = \begin{pmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & -2 \end{pmatrix}$. [MATH 247 | L-2/T-2 | 2020-21] **(18.67 marks)**

Q115. Determine whether the following polynomials are linearly independent or dependent: $P_1 = 1-x+2x^2$, $P_2 = 3+x$, $P_3 = 5-x+4x^2$, $P_4 = -2 - 2x + 2x^2$. [MATH 247 | L-2/T-2 | 2020-21] **(12 marks)** [MATH 259 | L-2/T-1 | 2013-14] **(10 marks)**

Q116. Define Vector space and Subspace with examples. Show that a subset of M_{mm} is a subspace and an invertible subset W of M_{mm} is not a subspace of M_{mm}. [MATH 247 | L-2/T-2 | 2020-21] **(10 marks)**

Q117. Determine whether the following polynomials span P_2: $p_1 = 1 - x - 2x^2$, $p_2 = 3 + x$, $p_3 = 5 - x + 4x^2$, $p_4 = -2 - 2x + 2x^2$. [MATH 247 | L-2/T-2 | 2020-21] **(10 marks)**

Q118. Let $S = \{v_1, v_2, v_3\}$ be a basis for $P_2(x)$ where $v_1 = 1 + 2x + x^2$, $v_2 = 2 + 9x$, $v_3 = 3 + 3x + 4x^2$. Find the coordinate vector of $p(x) = -1 + 18x + 12x^2$ relative to S. [MATH 247 | L-2/T-2 | 2020-21] **(10 marks)**

Q119. Given $A = \begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{pmatrix}$. Find bases for the row space and null space of A. Hence verify the dimension theorem. [MATH 247 | L-2/T-2 | 2019-20] **(20 marks)**

Q120. Determine whether the following subsets are subspaces of $\mathbb{R}^4$. If so, find a basis and dimension: (i) Set of vectors (a, b, c, d), where $b + c + d = 0$. (ii) Set of vectors (a, b, c, d), where $d = 2a + 7c$ and $3c = 2a - 5b$. [MATH 259 | L-2/T-1 | 2018-19] **(15 marks)**

Q121. Let $u = (2, 1, 0, 3)$, $v = (3, -1, 5, 2)$, $w = (-1, 0, 2, 1)$. Examine whether $x = (-4, 6, -13, 4)$ is in span$\{u, v, w\}$. [MATH 259 | L-2/T-1 | 2018-19] **(10 marks)**

Q122. Consider the basis $S = \{p_1, p_2, p_3\}$ for P_2 where $p_1 = 1 + x$, $p_2 = 1 + x^2$, $p_3 = x + x^2$. Is S a basis for P_2? If so, find the coordinate vector of $p(x) = 2 - x + x^2$ relative to S. [MATH 259 | L-2/T-1 | 2018-19] **(15 marks)**

↪ *Similar: MATH 215 | 2016-17*

Q123. Determine whether the set of all vectors of the form (a, b, c), where $b = a - c$, is a subspace of $\mathbb{R}^3$. [MATH 247 | L-2/T-2 | 2018-19] **(10 marks)**

Q124. Find the rank of the matrix $A = \begin{pmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2017-18] **(8 marks)**

Q125. Consider the basis $B = \{u_1, u_2\}$ and $B' = \{u_1', u_2'\}$ for $\mathbb{R}^2$, where $u_1 = (1,0)$, $u_2 = (0,1)$, $u_1' = (1,2)$, $u_2' = (2,1)$. (i) Find the transition matrix $P_{B' \to B}$. (ii) Find the transition matrix $P_{B \to B'}$. Use the formula to find $[v]_B$ given $[v]_{B'} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2017-18] **(15 marks)**

↪ *Similar: MATH 259 | 2021-22*

Q126. Find a subset of the vectors $v_1 = \begin{bmatrix} 1 & -1 \\ 5 & 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} -2 & 3 \\ 1 & 0 \end{bmatrix}$, $v_3 = \begin{bmatrix} 4 & -5 \\ 9 & 4 \end{bmatrix}$, $v_4 = \begin{bmatrix} 0 & 4 \\ 2 & -3 \end{bmatrix}$, $v_5 = \begin{bmatrix} -7 & 18 \\ 2 & -8 \end{bmatrix}$ that forms a basis for the space spanned by these vectors. Express each vector not in the basis as a linear combination of the basis vectors. Also find their co-ordinate vectors with respect to that basis. [MATH 259 | L-2/T-1 | 2017–18] **(20 marks)**

Q127. Determine a basis and dimension of the solution space for the homogeneous system: $2x_1+2x_2-x_3+x_5 = 0$, $-x_1-x_2+2x_3-3x_4+x_5 = 0$, $x_1 + x_2 - 2x_3 - x_5 = 0$, $x_3 + x_4 + x_5 = 0$. [MATH 247 | L-2/T-2 | 2017-18] **(16.67 marks)**

Q128. Find rank and nullity of the matrix $A = \begin{pmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{pmatrix}$. Hence verify the Dimension Theorem. [MATH 247 | L-2/T-2 | 2017-18] **(10 marks)**

Q129. If $u_1 = (1,0,2)$, $u_2 = (-1,1,0)$, $u_3 = (0,2,3)$ form a basis of $\mathbb{R}^3$, find the coordinates of $v = (2,5,11)$ relative to this basis. [MATH 247 | L-2/T-2 | 2017-18] **(10 marks)**

Q130. Let V be a vector space of all 2×2 matrices over the real field $\mathbb{R}$. Determine whether W is a subspace of V: (i) W consists of all non-singular matrices; (ii) W consists of all singular matrices. [MATH 247 | 2017–18]

Q131. Determine whether the subset $W = \{$all vectors of the form (a,b,c,d), where $d = a+b$ and $c = a-b\}$ is a subspace of $\mathbb{R}^4$. If so, find the dimension. [MATH 259 | L-2/T-1 | 2016-17] **(15 marks)**

Q132. Discuss how the rank of $A = \begin{pmatrix} t & 3 & -1 \\ 3 & 6 & -2 \\ -1 & -3 & t \end{pmatrix}$ varies with t. [MATH 259 | L-2/T-1 | 2016-17] **(15 marks)**

Q133. Let $S = \{(x,y,z,t) \mid y - 2z + t = 0\}$ and $T = \{(x,y,z,t) \mid x = t = 0,\ y - 2z = 0\}$. Find a basis and dimension of $S \cap T$. [MATH 247 | L-2/T-2 | 2016-17] **(15 marks)**

Q134. Let $W = \{(a,b,c) \mid a \geq b\}$ be a subset of $\mathbb{R}^3$. Test whether W is a subspace or not. [MATH 247 | L-2/T-2 | 2016-17] **(10 marks)**

Q135. Prove that the points (x_1,y_1), (x_2,y_2), (x_3,y_3) are collinear if and only if the rank of the matrix $A = \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}$ is less than 3. [MATH 259 | L-2/T-1 | 2015-16] **(8 marks)**

Q136. Determine whether the set $W = \{$all polynomials $a_0 + a_1x + a_2x^2 + a_3x^3$ for which $a_0 + a_1 + a_2 + a_3 = 0\}$ is a subspace of $P_3(x)$. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**

Q137. Determine whether the following set of vectors is a basis for $P_3(x)$: $p_1 = -1 + x$, $p_2 = 1 + x$, $p_3 = x^2$, $p_4 = x^3$. If so, find the coordinate vector of $p = 2 - x^2 + 3x^3$ relative to S. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q138. Find a basis for the row space of $A = \begin{pmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{pmatrix}$ consisting entirely of row vectors from A. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**

Q139. Determine whether the following are subspaces of M_{nn}: (i) the set of all $n \times n$ matrices A such that $A^T = -A$; (ii) the set of all $n \times n$ matrices A such that $\operatorname{tr}(A) = 0$. [MATH 259 | L-2/T-1 | 2014-15] **(12 marks)**

↪ *Similar: MATH 259 | 2017-18*

Q140. Find three vectors in $\mathbb{R}^3$ which are linearly dependent, and are such that any two of them are linearly independent. [MATH 259 | L-2/T-1 | 2014-15] **(10 marks)**

Q141. Find standard basis vectors for $\mathbb{R}^4$ that can be added to $\{v_1, v_2\}$ to produce a basis for $\mathbb{R}^4$, where $v_1 = (1,-4,2,-3)$ and $v_2 = (-3,8,-4,6)$. [MATH 243 | L-2/T-2 | 2014-15] **(15 marks)**

Q142. Examine whether the following are subspaces of $\mathbb{R}^3$: (i) $S = \{(x,y,z) : 3x + 2y - z = 0\}$; (ii) $S = \{(x,y,z) : x^2 \geq 1\}$. [MATH 243 | L-2/T-2 | 2014-15] **(10 marks)**

Q143. The vectors $u_1 = (1,2,0)$, $u_2 = (1,3,2)$, $u_3 = (0,1,3)$ form a basis S of $\mathbb{R}^3$. Find the coordinate vector of $v = (2,1,4)$ relative to S. [MATH 243 | L-2/T-2 | 2014-15] **(10 marks)**

Q144. Find basis for the column vectors of A and null space of A^T. Also verify the dimension theorem for: $A = \begin{pmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$.

[MATH 259 | 2014–15 (BUET)]

Q145. Determine whether the following set of vectors $A_1 = \begin{pmatrix} 3 & 6 \\ 3 & -6 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$, $A_3 = \begin{pmatrix} 0 & -8 \\ -12 & -4 \end{pmatrix}$, $A_4 = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$ can form a basis for M_{22}. If so, find the coordinate of $A = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$ relative to the basis $S = \{A_1, A_2, A_3, A_4\}$. Also find the vector $B \in M_{22}$ whose coordinate vector with respect to the basis S is $(B)_S = (-1,2,-3,4)$. [MATH 259 | L-2/T-1 | 2013-14] **(15 marks)**

Q146. Find a subset of the vectors $v_1 = (1, -1, 5, 2)$, $v_2 = (-2, 3, 1, 0)$, $v_3 = (4, -5, 9, 4)$, $v_4 = (0, 4, 2, -3)$, $v_5 = (-7, 18, 2, -8)$ that forms a basis for the space spanned by these vectors; then express each non-basis vector as a linear combination of the basis vectors. [MATH 259 | L-2/T-1 | 2013-14] **(15 marks)**

↪ *Similar: MATH 259 | 2016-17, 2020-21, 2021-22; MATH 215 | 2016-17; MATH 143 | 2022-23*

Q147. Find basis for the column space of $A = \begin{pmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{pmatrix}$ and the null space of A^T. Verify that every vector in the column space of A is orthogonal to every vector in the null space of A^T. [MATH 259 | L-2/T-1 | 2013-14] **(15 marks)**

Q148. Find basis for the column space of A and the null space of A^T. Verify the dimension theorem for $A = \begin{pmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{pmatrix}$.

[MATH 259 | L-2/T-1 |] **(15 marks)**

Q149. Let W be a subspace of $\mathbb{R}^5$ spanned by $u_1 = (1, 2, 1, 3, 2)$, $u_2 = (1, 3, 3, 5, 3)$, $u_3 = (3, 8, 7, 13, 8)$, $u_4 = (1, 4, 6, 9, 7)$, $u_5 = (5, 13, 13, 25, 19)$. Find a basis of W using original vectors, and find $\dim W$. [MATH 243 | L-2/T-2 | 2012-13] **(15 marks)**

Q150. Reduce the matrix $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 7 & 3 & 5 \\ 3 & 8 & 1 & -2 \\ 2 & 4 & 6 & 8 \end{pmatrix}$ to canonical form. Hence find the rank. [MATH 243 | L-2/T-2 | 2011-12] **(16.67 marks)**

Q151. Find a subset of vectors $v_1 = (1, -2, 0, 3)$, $v_2 = (2, -5, -3, 6)$, $v_3 = (1, -1, 3, 1)$, $v_4 = (2, -1, 4, -7)$, $v_5 = (3, 2, 14, -17)$ that forms a basis for their span; express non-basis vectors as linear combinations. [MATH 243 | L-2/T-2 | 2010-11] **(20.67 marks)**

↪ *Similar: MATH 143 | 2021-22*

Q152. Consider the two bases of $\mathbb{R}^3$: $E = \{e_1, e_2, e_3\}$ and $S = \{u_1, u_2, u_3\}$ where $u_1 = (1, 0, 1)$, $u_2 = (2, 1, 2)$, $u_3 = (1, 2, 2)$. (i) Find transition matrices P from E to S and Q from S to E. (ii) If $v = (1, 3, 5)$, find $[v]_S$ using the transition matrix. [MATH 243 | L-2/T-2 | 2010-11] **(26 marks)**

Q153. Find the coordinate vector of $v = (3, 1, -4)$ relative to the basis $\{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ of $\mathbb{R}^3$. [MATH 243 | L-2/T-2 | 2010-11] **(10 marks)**

Q154. Find a basis for the row space of $A = \begin{pmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{pmatrix}$ consisting entirely of row vectors from A. [MATH 243 | L-2/T-2 | 2010-11] **(15 marks)**

Q155. Consider the following two bases of $\mathbb{R}^3$: $E = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $S = \{(1, 0, 1), (2, 1, 2), (1, 2, 2)\}$. (i) Find the transition matrices P from E to S and Q from S to E. (ii) If $v = (1, 3, 5)$, find $[v]_S$ using the transition matrix. [MATH 243 | 2010–11]

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Linear Algebra & Matrix Theory

Orthogonality

Gram-Schmidt, QR Decomposition, Least Squares, Inner Products

S4 Orthogonality

Q156. Find the basis for the orthogonal complements of the row space of $A = \begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{pmatrix}$. Also verify the Dimension Theorem for A. [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

↪ *Similar: MATH 259 | 2016-17, 2020-21, 2021-22; MATH 215 | 2016-17; MATH 247 | 2019-20*

Q157. Discuss the Gram-Schmidt process. Let $\mathbb{R}^4$ have the Euclidean inner product. Transform the basis $\{u_1, u_2, u_3, u_4\}$, where $u_1 = (0, 2, 1, 0)$, $u_2 = (1, -1, 0, 0)$, $u_3 = (1, 2, 0, -1)$, $u_4 = (1, 0, 0, 1)$, into an orthogonal basis $\{v_1, v_2, v_3, v_4\}$; then normalize to obtain an orthonormal basis $\{q_1, q_2, q_3, q_4\}$. [MATH 143 | L-1/T-2 | 2023-24] **(20 marks)**

↪ *Similar: MATH 259 | 2013-14, 2017-18*

Q158. Define the least squares solution, the least squares error vector, and the least squares error of a linear system. [MATH 143 | L-1/T-2 | 2023-24] **(5 marks)**

↪ *Similar: MATH 143 | 2022-23*

Q159. Find the least squares solution, the least squares error vector, and the least squares error of the system $x - 3y - 2z = -7$, $2x + y + 3z = 0$, $y + z = 7$. Verify that (i) the error vector is orthogonal to the column space, and (ii) all solutions have the same error vector. [MATH 143 | L-1/T-2 | 2023-24] **(17 marks)**

Q160. Define orthogonal complement and the Gram-Schmidt process. [MATH 143 | L-1/T-2 | 2022-23] **(5 marks)**

Q161. Apply Gram-Schmidt process to transform $u_1 = (1, -1, 0)$, $u_2 = (2, 0, -2)$, $u_3 = (3, -3, 3)$ into an orthogonal basis; normalize to obtain the orthonormal basis; write the QR-decomposition of A whose columns are the given vectors. [MATH 143 | L-1/T-2 | 2022-23] **(15 marks)**

Q162. Find the least squares solutions, the least squares error vector, and the least squares error of: $3x + 2y - z = 2$, $x - 4y + 3z = -2$, $x + 10y - 7z = 1$. Verify that the error vector is orthogonal to the column space. [MATH 143 | L-1/T-2 | 2022-23] **(17 marks)**

Q163. Find a singular value decomposition (SVD) of the matrix $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2022-23] **(15 marks)**

Q164. Consider the vector space $\mathbb{R}^3$ with the Euclidean inner product. Apply Gram-Schmidt process to transform $u_1 = (1, 1, 1)$, $u_2 = (0, 1, 1)$, $u_3 = (0, 0, 1)$ into an orthogonal basis $\{v_1, v_2, v_3\}$ with $v_1 = u_1$; then normalize to obtain the orthonormal basis $\{q_1, q_2, q_3\}$. [MATH 143 | L-1/T-2 | 2021-22] **(18 marks)**

↪ *Similar: MATH 247 | 2020-21; MATH 215 | 2018-19*

Q165. Find the least squares solutions, the least squares error vector, and the least squares error of: $3x_1 + 2x_2 - x_3 = 2$, $x_1 - 4x_2 + 3x_3 = -2$, $x_1 + 10x_2 - 7x_3 = 1$. [MATH 143 | L-1/T-2 | 2021-22] **(18 marks)**

↪ *Similar: MATH 143 | 2022-23*

Q166. Determine whether $v_1 = (0, 1, -4, -1)$, $v_2 = (3, 5, 1, 1)$, $v_3 = (1, 0, 1, -4)$ are orthogonal with respect to the Euclidean inner product. If so, find $\mathrm{Proj}_W x$, where $x = (1, 2, 0, -2)$. [MATH 259 | L-2/T-1 | 2020-21] **(15 marks)**

Q167. Consider the vector space $\mathbb{R}^3$. Apply Gram-Schmidt process to transform $u_1 = (1, -1, 0)$, $u_2 = (2, 0, -2)$, $u_3 = (3, -3, 3)$ into an orthogonal basis; obtain the orthonormal basis; write the QR-decomposition of $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & -3 \\ 0 & -2 & 3 \end{pmatrix}$. [MATH 247 | L-2/T-2 | 2019-20] **(20 marks)**

Q168. Let W be the subspace of $\mathbb{R}^4$ spanned by $v_1 = (1, 4, 5, 2)$, $v_2 = (2, 1, 3, 0)$, $v_3 = (-1, 3, 2, 2)$. Find a basis for the orthogonal complement of W. [MATH 259 | L-2/T-1 | 2018-19] **(10 marks)**

Q169. Apply Gram-Schmidt process to transform $u_1 = (1, -1, 0)$, $u_2 = (2, 0, -2)$, $u_3 = (3, -3, 3)$ into an orthogonal basis; then obtain the orthonormal basis. Write the QR-decomposition of $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & -3 \\ 0 & -2 & 3 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2018-19] **(20 marks)**

↪ *Similar: MATH 247 | 2019-20; MATH 143 | 2022-23*

Q170. Consider the vector space $\mathbb{R}^3$ with the Euclidean inner product. Apply Gram-Schmidt process to transform $u_1 = (1, 1, 1)$, $u_2 = (0, 1, 1)$, $u_3 = (0, 0, 1)$ into an orthogonal basis with $v_1 = u_1$; normalize to get orthonormal basis; find the QR-decomposition of $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$. [MATH 215 | L-2/T-2 | 2018-19] **(15 marks)**

Q171. Use Gram-Schmidt process to transform the basis $B = \{(4, -3, 0), (1, 2, 0), (0, 0, 4)\}$ into an orthogonal basis for $\mathbb{R}^3$. Also find the QR-decomposition of $A = \begin{pmatrix} 4 & 1 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$. [MATH 215 | L-2/T-2 | 2017-18] **(20 marks)**

Q172. Let $\mathbb{R}^4$ have inner product $\langle u, v\rangle = u_1v_1 + 2u_2v_2 + 3u_3v_3 + 4u_4v_4$. Use Gram-Schmidt to transform the basis $u_1 = (0, 2, 1, 0)$, $u_2 = (1, -1, 0, 0)$, $u_3 = (1, 2, 0, -1)$, $u_4 = (1, 0, 0, 1)$ into an orthogonal basis; also find the angle between u_1 and u_2. [MATH 259 | L-2/T-1 | 2017-18] **(20 marks)**

Q173. Let W be the subspace of $\mathbb{R}^6$ spanned by $w_1 = (1, 3, -2, 0, 2, 0)$, $w_2 = (2, 6, -5, -2, 4, -3)$, $w_3 = (0, 0, 5, 10, 0, 15)$, $w_4 = (2, 6, 0, 8, 4, 18)$. Find a basis for the orthogonal complement of W. [MATH 259 | L-2/T-1 | 2017-18] **(15 marks)**

Q174. Find a basis for the orthogonal complement of the subspace of $\mathbb{R}^3$ spanned by $v_1 = (1, -1, 3)$, $v_2 = (5, -4, -4)$, $v_3 = (7, -6, 2)$. [MATH 215 | L-2/T-2 | 2017-18] **(10 marks)**

Q175. Let P_2 have the inner product $\langle u, v\rangle = \int_{-1}^{1} u(x)v(x)\,dx$. Use the Gram-Schmidt process to transform the basis $S = \{1 + x,\ 1 + x^2,\ x + x^2\}$ into an orthonormal basis $\{q_1, q_2, q_3\}$. Hence find the coordinate vector of $w = 2 - x + x^2$ relative to this basis. Also verify the Pythagorean theorem for q_1 and q_3. [MATH 259 | L-2/T-1 | 2016-17] **(35 marks)**

Q176. Orthonormalize the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 1, 0)$, $u_3 = (1, 0, 0)$ using the Gram-Schmidt process and verify orthonormality. [MATH 247 | L-2/T-2 | 2016-17] **(23.33 marks)**

↪ *Similar: MATH 247 | 2017-18, 2018-19, 2021-22*

Q177. Apply Gram-Schmidt process to transform $u_1 = (0, 1, 2)$, $u_2 = (-1, 0, 1)$, $u_3 = (-1, 1, 3)$ into an orthogonal basis; normalize to obtain the orthonormal basis; write the QR-decomposition of $A = \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{pmatrix}$. [MATH 215 | L-2/T-2 | 2016-17] **(17 marks)**

Q178. Let W be the subspace of $\mathbb{R}^5$ spanned by $w_1 = (2, 2, -1, 0, 1)$, $w_2 = (-1, -1, 2, -3, 1)$, $w_3 = (1, 1, -2, 0, -1)$, $w_4 = (0, 0, 1, 1, 1)$. Find a basis for the orthogonal complement of W. [MATH 259 | L-2/T-1 | 2015-16] **(20 marks)**

Q179. Find the QR-decomposition of the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}$, if possible. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q180. Let $\mathbb{R}^4$ have the inner product $\langle u, v\rangle = u_1v_1 + 2u_2v_2 + 3u_3v_3 + 4u_4v_4$. Use the Gram-Schmidt process to transform the basis $u_1 = (0, 2, 1, 0)$, $u_2 = (1, -1, 0, 0)$, $u_3 = (1, 2, 0, -1)$, $u_4 = (1, 0, 0, 1)$ into an orthonormal basis. [MATH 259 | L-2/T-1 | 2013-14] **(20 marks)**

↪ *Similar: MATH 259 | 2017-18*

BUET · Linear Algebra & Matrix Theory

Linear Algebra & Matrix Theory

Eigenvalues & Eigenvectors

Diagonalization, Cayley-Hamilton, Minimal Polynomial, Quadratic Forms

S5 Eigenvalues & Eigenvectors

Q181. Define diagonalizability of a square matrix. Find the characteristic roots and characteristic vectors of $A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$. Find the matrix S that diagonalizes A, determine $S^{-1}AS$, and compute A^{13}. [MATH 143 | L-1/T-2 | 2023-24] **(20 marks)**

↪ *Similar: MATH 259 | 2017-18, 2019-20*

Q182. Identify the conic section represented by $11x^2 + 24xy + 4y^2 - 15 = 0$ by rotating axes to standard position. Find the equation in rotated coordinates and the angle of rotation. [MATH 143 | L-1/T-2 | 2023-24] **(13 marks)**

Q183. Define positive definite, negative definite, and indefinite quadratic form. Classify the matrix $A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 3 & 2 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2023-24] **(10 marks)**

↪ *Similar: MATH 143 | 2021-22*

Q184. What is the condition for a matrix to be diagonalizable? Determine whether $A = \begin{pmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{pmatrix}$ is diagonalizable. If so, find P and $P^{-1}AP$. Also compute A^{11}. [MATH 143 | L-1/T-2 | 2022-23] **(20 marks)**

Q185. Using eigenvalues, classify the matrix $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ as positive definite, negative definite, or indefinite. [MATH 143 | L-1/T-2 | 2022-23] **(6 marks)**

Q186. Identify the conic whose equation is $5x^2 - 4xy + 8y^2 - 36 = 0$ by rotating the xy-axes to standard position. [MATH 143 | L-1/T-2 | 2022-23] **(12 marks)**

Q187. What is the condition of a matrix being diagonalizable? Determine whether $A = \begin{pmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{pmatrix}$ is diagonalizable. If so, find P, $P^{-1}AP$, and compute A^{11}. [MATH 143 | 2022–23]

Q188. Find the eigenspaces of $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2021-22] **(18 marks)**

Q189. Determine whether $A = \begin{pmatrix} 5 & 4 & -1 \\ 4 & 5 & -1 \\ -4 & -4 & 2 \end{pmatrix}$ is diagonalizable. [MATH 259 | L-2/T-1 | 2021-22] **(17 marks)**

Q190. Find a matrix P that diagonalizes $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$ and verify $P^{-1}AP$ is diagonal. Also find A^{13}. [MATH 143 | L-1/T-2 | 2021-22] **(20 marks)**

Q191. (i) Write down the system $y_1' = y_1 + 4y_2$, $y_2' = 2y_1 + 3y_2$ in matrix form. (ii) Use diagonalization to solve the system with initial conditions $y_1(0) = 0$, $y_2(0) = 0$. [MATH 143 | L-1/T-2 | 2021-22] **(15 marks)**

Q192. (i) Explain principal sub-matrix. Using principal sub-matrices, classify $A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 3 & 2 \end{pmatrix}$ as positive definite, negative definite, or indefinite. (ii) Identify all values of k for which $5x_1^2 + x_2^2 + kx_3^2 + 4x_1x_2 - 2x_1x_3 - 2x_2x_3$ is positive definite. [MATH 143 | L-1/T-2 | 2021-22] **(17 marks)**

Q193. Write down the symmetric matrix A corresponding to the quadratic form $q = x^2+7y^2+8z^2-6xy+4xz-10yz$. Find a nonsingular matrix P such that $P^TAP = D$. Write the rank, index, and signature. Identify the geometrical object represented by $q = \text{const}$. [MATH 259 | L-2/T-1 | 2021-22] **(15 marks)**

Q194. Write down the system $y_1' = y_1+4y_2$, $y_2' = 2y_1+3y_2$ in matrix form. Use diagonalization to solve with initial conditions $y_1(0) = 0$, $y_2(0) = 0$. [MATH 143 | 2021–22]

Q195. Explain principal sub-matrix. Working with principal sub-matrices, classify the matrix $A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 3 & 2 \end{pmatrix}$ as positive definite, negative definite, or indefinite. Also identify all values of k for which $5x_1^2+x_2^2+kx_3^2+4x_1x_2-2x_1x_3-2x_2x_3$ is positive definite. [MATH 143 | 2021–22]

Q196. Rectify the following statements: (i) For every square matrix A there exists a matrix P such that $P^{-1}AP$ is diagonal. (ii) Diagonalizability of a square matrix indicates the complete distinction of its eigenvalues. [MATH 259 | L-2/T-1 | 2020-21] **(10 marks)**

Q197. Consider the quadratic form $Q(x) = 21x_1^2 + 11x_2^2 + 2x_3^2 - 30x_1x_2 - 8x_2x_3 + 12x_3x_1$. Find a non-singular linear transformation to reduce it to canonical form. Find rank, index, and signature. Comment on possible outputs and give a non-trivial example for each. [MATH 259 | L-2/T-1 | 2020-21] **(18 marks)**

Q198. Determine the minimal polynomial of $\begin{pmatrix}1&1&0\\0&2&0\\0&0&1\end{pmatrix}$. [MATH 259 | L-2/T-1 | 2020-21] **(7 marks)**

Q199. If $A=\begin{pmatrix}4&0&1\\2&3&2\\1&0&4\end{pmatrix}$: (i) Find the eigenvalues of A. (ii) For each eigenvalue, find the eigenvectors. (iii) Find the rank of $\lambda I-A$. (iv) Is A diagonalizable? Find P. (v) Find $P^{-1}AP$ and A^{100}. [MATH 247 | L-2/T-2 | 2020-21] **(26.67 marks)**

Q200. Use the Cayley-Hamilton theorem to calculate A^n where $A=\begin{pmatrix}1&3\\4&7\end{pmatrix}$ and n is any positive integer. [MATH 259 | L-2/T-1 | 2020-21] **(10 marks)**

Q201. Rectify the following statements with all possible options along with the reasonings: (i) For every square matrix A there exists a matrix P such that $P^{-1}AP$ is diagonal. (ii) Diagonalizability of a square matrix indicates the complete distinction of its eigenvalues. [MATH 259 | 2020–21 (BUET)]

Q202. Find a matrix P that diagonalizes the matrix $C=\begin{pmatrix}1&2&3\\2&4&7\\3&6&10\end{pmatrix}$. [MATH 259 | 2020–21 (BUET)]

Q203. If $A=\begin{pmatrix}4&0&1\\2&3&2\\1&0&4\end{pmatrix}$: (i) find the eigenvalues; (ii) for each eigenvalue find the eigenvectors; (iii) find the rank of $\lambda I-A$; (iv) is A diagonalizable? If so, find P; (v) find $P^{-1}AP$ and A^{100}. [MATH 247 | 2020–21]

Q204. Find a matrix P that diagonalizes $C=\begin{pmatrix}1&2&3\\2&4&7\\3&6&10\end{pmatrix}$. [MATH 259 | 2020–21 (BUET)]

Q205. Find the eigenvalues and corresponding eigenvectors of $A=\begin{pmatrix}1&-3&3\\3&-5&3\\6&-6&4\end{pmatrix}$. Construct the eigenvector matrix S that diagonalizes A and determine the eigenvalue matrix Λ. [MATH 247 | L-2/T-2 | 2019-20] **(26.67 marks)**

↪ *Similar: MATH 215 | 2017-18*

Q206. Prove that if λ is an eigenvalue of an invertible matrix A, then $1/\lambda$ is an eigenvalue of A^{-1}. [MATH 259 | L-2/T-1 | 2018-19] **(11 marks)**

Q207. Determine whether $A=\begin{pmatrix}1&3&4\\3&5&6\\4&6&7\end{pmatrix}$ is diagonalizable. If so, find P such that $P^{-1}AP=D$. Also compute A^{17}. [MATH 259 | L-2/T-1 | 2018-19] **(12 marks)**

↪ *Similar: MATH 259 | 2018-19*

Q208. State Cayley-Hamilton theorem. Verify it for $A=\begin{pmatrix}2&2&-2\\2&3&-1\\-2&-1&3\end{pmatrix}$ and hence find A^{-1}. [MATH 259 | L-2/T-1 | 2018-19] **(20 marks)**

Q209. Reduce the quadratic form $q=x_1^2+2x_2^2-3x_3^2+8x_1x_2+10x_1x_3-16x_2x_3$ to canonical form. Find the linear transformation, rank, index, and signature. [MATH 259 | L-2/T-1 | 2018-19] **(15 marks)**

↪ *Similar: MATH 259 | 2017-18; MATH 215 | 2018-19*

Q210. Find all eigenvalues and corresponding eigenvectors of $A=\begin{pmatrix}3&2&4\\2&0&2\\4&2&3\end{pmatrix}$. [MATH 247 | L-2/T-2 | 2018-19] **(20 marks)**

Q211. Find the inverse of the matrix $A=\begin{pmatrix}1&2&3\\2&-1&1\\3&4&1\end{pmatrix}$ by Cayley-Hamilton theorem. [MATH 247 | L-2/T-2 | 2018-19] **(20 marks)**

Q212. Find all eigenvalues and corresponding eigenvectors of $A=\begin{pmatrix}1&0&0\\3&-5&0\\6&-6&4\end{pmatrix}$. [MATH 215 | L-2/T-2 | 2018-19] **(15 marks)**

Q213. Determine whether $A=\begin{pmatrix}1&3&4\\3&5&6\\4&6&7\end{pmatrix}$ is diagonalizable. If so, find P such that $P^{-1}AP=D$. Compute A^{17}. [MATH 259 | 2018–19, 2021–22 (BUET)]

Q214. If $\lambda_1,\dots,\lambda_n$ are the eigenvalues of A, prove that $\sum\lambda_i=\operatorname{tr}(A)$ and $\prod\lambda_i=\det(A)$. [MATH 259 | L-2/T-1 | 2017-18] **(8 marks)**

Q215. Find eigenvalues and eigenvectors of $A=\begin{pmatrix}2&2&1\\1&3&1\\1&2&2\end{pmatrix}$. Is A diagonalizable? [MATH 259 | L-2/T-1 | 2017-18] **(12 marks)**

Q216. Verify Cayley-Hamilton theorem for $A=\begin{pmatrix}1&0&3\\2&1&-1\\1&-1&1\end{pmatrix}$ and express A^5 as a linear combination of A^2, A, and I. [MATH 259 | L-2/T-1 | 2017-18] **(15 marks)**

Q217. Determine whether $A = \begin{pmatrix} -1 & -3 & -3 \\ 4 & 4 & 1 \\ -2 & 0 & 3 \end{pmatrix}$ is diagonalizable or not. If so, find P, $P^{-1}AP$, and compute A^{21}. [MATH 247 | L-2/T-2 | 2017-18] **(23.33 marks)**

Q218. Find the minimal polynomial of the matrix $A = \begin{pmatrix} 5 & -1 & 0 \\ -1 & 5 & 0 \\ 0 & 0 & -2 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2017-18] **(10 marks)**

Q219. Define a real quadratic form. Reduce $Q = x_1^2 + 2x_2^2 - 2x_3^2 + 4x_1x_2 + 6x_1x_3$ to canonical form. Find rank, index, and signature. Write the equations of transformation. [MATH 215 | L-2/T-2 | 2017-18] **(15 marks)**

Q220. Determine whether the following matrix A is diagonalizable. If so, find a matrix P that diagonalizes $A = \begin{pmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{pmatrix}$, find $P^{-1}AP$, and compute A^{21}. [MATH 247 | 2017–18]

Q221. Find the eigenspace, algebraic multiplicity, and geometric multiplicity for each eigenvalue of $A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2016-17] **(18 marks)**

Q222. Find the degree of the minimal polynomial of $A = \begin{pmatrix} 6 & 2 & -2 \\ 2 & 3 & -1 \\ -2 & -1 & 3 \end{pmatrix}$. Is A derogatory? If yes, express the characteristic polynomial as the product of the minimal polynomial and a monic factor. [MATH 259 | L-2/T-1 | 2016-17] **(15 marks)**
↪ *Similar: MATH 259 | 2020-21*

Q223. Write the condition for positive and negative definite quadratic forms. Reduce the quadratic form $q = 4x_1^2 + 3x_2^2 - x_3^2 + 2x_2x_3 - 4x_3x_1 + 4x_1x_2$ to canonical form. Find rank, index, and signature. Write the equations of transformation. [MATH 215 | L-2/T-2 | 2016-17] **(18 marks)**

Q224. Check whether $B = \begin{pmatrix} 6 & 2 & -2 \\ 2 & 3 & -1 \\ -2 & -1 & 3 \end{pmatrix}$ is derogatory or not. If so, express the characteristic polynomial as the product of the minimal polynomial and a monic factor. [MATH 215 | L-2/T-2 | 2016-17] **(18 marks)**

Q225. Prove that $\mathbf{u}$ is an eigenvector of $A^{n\times n}$ with eigenvalue λ. Then prove that the eigenvalue of $a_kA^k + a_{k-1}A^{k-1} + \cdots + a_1A + a_0I$ associated with eigenvector $\mathbf{u}$ is $a_k\lambda^k + \cdots + a_1\lambda + a_0$. [MATH 259 | L-2/T-1 | 2016-17] **(10 marks)**

Q226. If $A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 2 & -2 \end{pmatrix}$, find the value of A^n using Cayley-Hamilton theorem. Verify your result for $n = 2$. [MATH 259 | L-2/T-1 | 2016-17] **(15 marks)**

Q227. Define quadratic form and explain its matrix representation. Reduce the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 0 \\ 3 & 0 & -2 \end{pmatrix}$ to diagonal form. Interpret the result in terms of quadratic form and write the equations of transformation. [MATH 259 | L-2/T-1 | 2016-17] **(15 marks)**

Q228. Find the characteristic polynomial, characteristic equation, eigenvalues, eigenvectors and eigenspace for the matrix $B = \begin{pmatrix} 5 & -1 & 1 \\ -1 & 2 & -4 \\ 1 & -4 & 2 \end{pmatrix}$. **(17 marks)** [MATH 215 | L2-T2 | 2016–17]

Q229. Find the degree of the minimal polynomial of $A = \begin{pmatrix} 6 & 2 & -2 \\ 2 & 3 & -1 \\ -2 & -1 & 3 \end{pmatrix}$. Is A derogatory? If so, express the characteristic polynomial as the product of the minimal polynomial and a monic factor. [MATH 259 | 2016–17 (BUET)]

Q230. Find a matrix P that diagonalizes $A = \begin{pmatrix} -1 & -2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{pmatrix}$ and find $P^{-1}AP$. [MATH 215 | 2016–17]

Q231. Diagonalise the matrix $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$, if possible. **(18 marks)** [MATH 215 | L2-T-2 | 2016–17]

Q232. If $A = \begin{pmatrix} 2 & 3 & 5 \\ 3 & 2 & 7 \\ 0 & 0 & 4 \end{pmatrix}$, find the eigenvalues of $3A^3 + 5A^2 - 6A + 2I - 8A^{-1}$. [MATH 259 | L-2/T-1 | 2015-16] **(12 marks)**

Q233. Diagonalize the matrix $A = \begin{pmatrix} 4 & 0 & 8 \\ 0 & 12 & 0 \\ 8 & 0 & 4 \end{pmatrix}$ by means of an orthogonal transformation. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q234. Find the minimal polynomial of $A = \begin{pmatrix} 3 & 5 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**

Q235. Using Cayley-Hamilton theorem, find the value of A^n for $A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**

↪ *Similar: MATH 259 | 2016-17, 2020-21*

Q236. Define quadratic form and explain its matrix representation. Deduce the quadratic form $q = x^2 + 2y^2 + 3z^2 - 2xy + 2yz$ into sum of squares. Find nature, rank, index, and signature. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q237. Find all eigenvalues, eigenvectors and the corresponding eigenspaces of the matrix $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}$. [MATH 259 | 2015–16]

Q238. State Cayley-Hamilton theorem and using this theorem find the inverse of $A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$. Also find A^{-2} and A^3. [MATH 259 | 2015–16]

Q239. Find the minimal and characteristic equations of $A = \begin{pmatrix} 5 & 4 & -1 \\ 4 & 5 & -1 \\ -4 & -4 & 3 \end{pmatrix}$. Is A derogatory? If yes, express the characteristic polynomial as the product of the minimal polynomial and a monic factor. [MATH 259 | L-2/T-1 | 2014-15] **(18 marks)**

Q240. Find the eigenvalues, eigenvectors, and characteristic space for $A = \begin{pmatrix} 5 & -1 & 1 \\ -1 & 2 & -4 \\ 1 & -4 & 2 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2014-15] **(17 marks)**

Q241. State and verify Cayley-Hamilton theorem for $B = \begin{pmatrix} 4 & -1 & 1 \\ -1 & 4 & -1 \\ 1 & -1 & 4 \end{pmatrix}$. Also find B^{-1}. [MATH 259 | L-2/T-1 | 2014-15] **(17 marks)**

↪ *Similar: MATH 259 | 2018-19*

Q242. Find the eigenvalues and corresponding eigenvectors of matrix $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$. Find a matrix P that diagonalizes A and find $P^{-1}AP$. [MATH 259 | 2014–15]

Q243. Verify Cayley-Hamilton theorem for the matrix $\begin{pmatrix} 2 & 7 & 0 \\ 4 & 5 & 1 \\ 3 & 2 & 1 \end{pmatrix}$. Also find the inverse using this theorem. [MATH 259 | L-2/T-1 | 2013-14] **(18 marks)**

Q244. What is a minimal polynomial? Find the minimal polynomial of $\begin{pmatrix} 3 & 2 & 1 \\ 4 & 5 & 1 \\ 7 & 8 & 0 \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2013-14] **(17 marks)**

Q245. Find the eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 5 & -1 & 1 \\ -1 & 2 & -4 \\ 1 & -4 & 2 \end{pmatrix}$. Construct a matrix P that diagonalizes A. [MATH 259 | L-2/T-1 | 2013-14] **(20 marks)**

Q246. Reduce the quadratic form $q = 4x_1^2 + 3x_2^2 - x_3^2 + 2x_2x_3 - 4x_3x_1 + 4x_1x_2$ to canonical form. Find the rank, index, and signature. Write the equations of transformation. [MATH 259 | L-2/T-1 | 2013-14] **(15 marks)**

↪ *Similar: MATH 259 | 2014-15, 2015-16, 2017-18, 2018-19; MATH 215 | 2016-17*

Q247. Construct (if possible) a matrix P that will diagonalize A and verify that $P^{-1}AP$ is a diagonal matrix, where $A = \begin{pmatrix} 5 & -1 & 1 \\ -1 & 2 & -4 \\ 1 & -4 & 2 \end{pmatrix}$. [MATH 259 | 2013–14]

Q248. Construct (if possible) a matrix P that will diagonalize A and verify that $P^{-1}AP$ is diagonal, where $A = \begin{pmatrix} 5 & -1 & 1 \\ -1 & 2 & -4 \\ 1 & -4 & 2 \end{pmatrix}$. [MATH 259 | 2013–14]

Q249. State Cayley-Hamilton theorem and verify it for the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 6 \\ 2 & 6 & 13 \end{pmatrix}$. Hence find the inverse of A. [MATH 243 | L-2/T-2 | 2012-13] **(16.67 marks)**

Q250. Find all eigenvalues and corresponding eigenvectors of $A = \begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{pmatrix}$. Find P that diagonalizes A and determine $P^{-1}AP$. [MATH 243 | L-2/T-2 | 2011-12] **(26.67 marks)**

Q251. Find all eigenvalues and the corresponding eigenvectors of the matrix
$A = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ -1 & -4 & -3 \end{pmatrix}$. Also find the matrix P that diagonalizes A. [MATH 243 | L-2/T-2 | 2011-12] **(20 marks)**

Q252. Use Cayley-Hamilton theorem to find the inverse of the matrix $A = \begin{pmatrix} 2 & 4 & 3 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$. [MATH 243 | L-2/T-2 | 2011-12] **(15 marks)**

Q253. State Cayley-Hamilton theorem and verify it for the matrix $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$. Hence find the inverse of A using the theorem. [MATH 243 | L-2/T-2 | 2010-11] **(16.67 marks)**

Q254. Find the eigenvalues and the corresponding eigenvectors of the matrix
$A = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ -4 & -4 & 2 \end{pmatrix}$. Is the matrix A diagonalizable? If so, write down the nonsingular matrix P that diagonalizes A and the corresponding diagonal matrix D. [MATH 243 | L-2/T-2 | 2010-11] **(20 marks)**

Q255. State Cayley-Hamilton theorem and verify it for the matrix $A = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ -4 & -4 & 2 \end{pmatrix}$. Hence find the inverse of A using the above theorem. [MATH 243 | L-2/T-2 | 2010-11] **(16.67 marks)**

Q256. Define semi-definite and indefinite real quadratic forms. Reduce the real quadratic form $q = 4x_1^2 + 3x_2^2 - x_3^2 + 2x_2x_3 - 4x_3x_1 + 4x_1x_2$ to canonical form. [MATH 259 | L-2/T-1 | Classic] **(12 marks)**

Q257. Determine whether the following matrix is diagonalizable. If so, find a nonsingular matrix P that diagonalizes $A = \begin{pmatrix} 5 & 4 & -1 \\ 4 & 5 & -1 \\ -4 & -4 & 2 \end{pmatrix}$ and write down the diagonal matrix D. [MATH 259 | Classic]

BUET · Linear Algebra & Matrix Theory

Linear Algebra & Matrix Theory

Linear Transformations

Kernel, Range, Standard Matrix, Composition, Matrix Representations

S6 Linear Transformations

Q258. Let $T : P_2 \to P_2$ be the linear operator defined by $T(p(x)) = p(3x-5)$. (i) Find the matrix $[T]_B$ relative to the basis $B = \{1, x, x^2\}$. (ii) Compute $T(1 + 2x + 3x^2)$. **[MATH 143 | L-1/T-2 | 2023-24] (15 marks)**

↪ *Similar: MATH 259 | 2017-18*

Q259. In $\mathbb{R}^3$, the shear in the xy-direction by factor k moves each point (x, y, z) to $(x + kz, y + kz, z)$. (i) Find the standard matrix for this shear. (ii) Define shears in the xz- and yz-directions and give their standard matrices. **[MATH 143 | L-1/T-2 | 2023-24] (15 marks)**

Q260. Let $T : P_2 \to P_3$ be the linear transformation defined by $T(p(x)) = x \cdot p(3x - 5)$. (i) Find $[T]_{B,B'}$ relative to $B = \{1, x, x^2\}$ and $B' = \{1, x, x^2, x^3\}$. (ii) Compute $T(1 + 2x + 3x^2)$. **[MATH 143 | L-1/T-2 | 2023-24] (10 marks)**

↪ *Similar: MATH 259 | 2018-19*

Q261. Determine whether $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + 2y - z,\ y + z,\ x + y - 2z)$ is a linear operator. Find basis and dimension of (i) $\operatorname{Im} T$ and (ii) $\ker T$. **[MATH 143 | L-1/T-2 | 2023-24] (15 marks)**

Q262. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by $T(x, y, s, t) = (4x + y - 2s - 3t,\ 2x + y + s - 4t,\ 6x - 9s + 9t)$. Find basis and dimension of (i) $\ker T$ and (ii) $\operatorname{Range} T$. **[MATH 143 | L-1/T-2 | 2022-23] (15 marks)**

Q263. List the standard matrices for the reflection operators about the xy-, xz-, and yz-planes, and the projection operators onto the xy-, xz-, and yz-planes. **[MATH 143 | L-1/T-2 | 2022-23] (5 marks)**

Q264. Find the standard matrix for T on $\mathbb{R}^3$, where T is a rotation of $30°$ about the y-axis, followed by a reflection about the xy-plane, followed by an orthogonal projection on the yz-plane. Find $T(2, -1, 4)$. **[MATH 143 | L-1/T-2 | 2022-23] (15 marks)**

Q265. Consider the basis $S = \{u, v, w\}$ for $\mathbb{R}^3$, where $u = (1, 1, 1)$, $v = (1, 1, 0)$, $w = (1, 0, 0)$. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ with $T(u) = (2, -1, 4)$, $T(v) = (3, 0, 1)$, $T(w) = (-1, 5, 1)$. Find a formula for $T(x, y, z)$ and use it to find $T(2, -3, 5)$. **[MATH 143 | L-1/T-2 | 2022-23] (15 marks)**

Q266. Determine whether $T : M_{33} \to \mathbb{R}$, where $T(A) = \det A$, is a linear transformation. **[MATH 259 | L-2/T-1 | 2021-22] (10 marks)**

Q267. Find the standard matrix for the transformation T on $\mathbb{R}^3$, where T is a rotation of $45°$ about the y-axis, followed by a reflection about the yz-plane, followed by a dilation with factor $k = \sqrt{2}$. Find $T(3, -4, 9)$. **[MATH 143 | L-1/T-2 | 2021-22] (17 marks)**

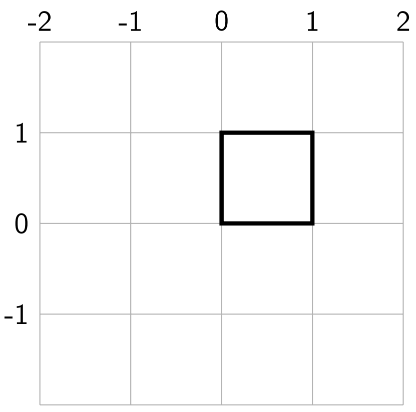
↪ *Similar: MATH 215 | 2016-17, 2017-18*

Q268. Find the Kernel and Range of the orthogonal projection on the plane $x = z$. Write down the basis and dimension. **[MATH 143 | L-1/T-2 | 2021-22] (18 marks)**

Q269. Consider the basis $S = \{v_1, v_2, v_3\}$ for $\mathbb{R}^3$, where $v_1 = (1, 2, 1)$, $v_2 = (2, 9, 0)$, $v_3 = (3, 3, 4)$. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ with $T(v_1) = (1, 0)$, $T(v_2) = (-1, 1)$, $T(v_3) = (0, 1)$. Find a formula for $T(x, y, z)$ and use it to find $T(7, 13, 7)$. **[MATH 143 | L-1/T-2 | 2021-22] (17 marks)**

↪ *Similar: MATH 247 | 2018-19*

Q270. (i) Write down the coordinate matrix of a square with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, $(0, 1, 0)$. (ii) Scale by $-\frac{3}{2}$ in x and $\frac{1}{2}$ in y. (iii) Rotate through $-60°$ about the z-axis. **[MATH 143 | L-1/T-2 | 2021-22] (13 marks)**



View 1: Square with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, and $(0, 1, 0)$.

Q271. Find the standard matrix for the transformation T on $\mathbb{R}^3$, where T is the composition of a rotation of $-30°$ about the x-axis, followed by a contraction with factor $\frac{1}{4}$, followed by a reflection about the yz-plane, followed by an orthogonal projection on the xz-plane. **[MATH 259 | L-2/T-1 | 2021-22] (15 marks)**

Q272. Describe the geometric effect of multiplying a vector x by $A = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$. **[MATH 259 | L-2/T-1 | 2020-21] (10 marks)**

Q273. Determine whether $T : M_{nn} \to \mathbb{R}$, where $T(A) = \operatorname{tr}(A)$, is a linear transformation. **[MATH 259 | L-2/T-1 | 2020-21] (10 marks)**

Q274. Describe the kernel and range of the orthogonal projection on the plane $y = x$. **[MATH 259 | L-2/T-1 | 2020-21] (10 marks)**

↪ *Similar: MATH 247 | 2018-19; MATH 143 | 2021-22*

Q275. Find the standard matrix for the composition T on $\mathbb{R}^3$: rotation of $60°$ about the z-axis, followed by a reflection about the xy-plane, followed by a contraction with factor $k = \frac{1}{2}$. Find the image of $(2, 1, -1)$ and explain geometrically. **[MATH 247 | L-2/T-2 | 2020-21] (16 marks)**

Q276. Let $T_1(x_1, x_2, x_3) = (4x_1, -2x_1 + x_2, -x_1 - 3x_2)$ and $T_2(x_1, x_2, x_3) = (x_1 + 2x_2, -x_3, 4x_1 - x_3)$. Find the standard matrices for T_1 and T_2. Also find $T_2 \circ T_1$ and $T_1 \circ T_2$. **[MATH 247 | L-2/T-2 | 2020-21] (14.67 marks)**

Q277. Find the standard matrix for T on $\mathbb{R}^3$, where T is a rotation of $60°$ about the y-axis, followed by a reflection about the yz-plane, followed by a dilation with factor $k = 2$. Find $T(2, -1, 4)$. [MATH 247 | L-2/T-2 | 2019-20] **(16.67 marks)**

Q278. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by $T(x, y, z, t) = (x - y + z + t,\ x + 2z - t,\ x + y + 3z - 3t)$. Find basis and dimension of Range(T) and ker(T). [MATH 247 | L-2/T-2 | 2019-20] **(16.67 marks)**

↪ *Similar: MATH 243 | 2010-11; MATH 247 | 2018-19*

Q279. Determine whether $T(x, y, s, t) = (4x + y - 2s - 3t,\ 2x + y + s - 4t,\ 6x - 9s + 9t)$ is a linear operator. Find basis and dimension of $\ker T$ and Range T. [MATH 259 | L-2/T-1 | 2018-19] **(20 marks)**

↪ *Similar: MATH 259 | 2014-15, 2021-22; MATH 143 | 2022-23*

Q280. Let $T : P_2 \to P_3$ be the linear transformation $T(p(x)) = xp(3x - 5)$. Find the matrix for T with respect to the standard bases $B = \{1, x, x^2\}$ and $B' = \{1, x, x^2, x^3\}$. Verify $[T]_{B',B}[x]_B = [T(x)]_{B'}$. [MATH 259 | L-2/T-1 | 2018-19] **(15 marks)**

↪ *Similar: MATH 215 | 2016-17*

Q281. Find the standard matrix for T on $\mathbb{R}^3$, where T is the composition of a reflection about the xy-plane, followed by a reflection about the xz-plane, followed by an orthogonal projection on the yz-plane. [MATH 259 | L-2/T-1 | 2018-19] **(17 marks)**

Q282. Determine whether the linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $w_1 = x_1 + 4x_2 - x_3$, $w_2 = 2x_1 + 7x_2 + x_3$, $w_3 = x_1 + 3x_2$ is one-to-one. If so, find the standard matrix for the inverse operator and find T^{-1}. [MATH 259 | L-2/T-1 | 2018-19] **(18 marks)**

Q283. Consider the basis $S = \{v_1, v_2, v_3\}$ for $\mathbb{R}^3$, where $v_1 = (1, 2, 1)$, $v_2 = (2, 9, 0)$, $v_3 = (3, 3, 4)$. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ with $T(v_1) = (1, 0)$, $T(v_2) = (-1, 1)$, $T(v_3) = (0, 1)$. Find a formula for $T(x, y, z)$ and compute $T(7, -3, 5)$. [MATH 247 | L-2/T-2 | 2018-19] **(20 marks)**

Q284. Let the linear transformation be defined by $T(x, y, z) = (x - y - 3z,\ 5x + 6y - 4z,\ 7x + 4y + 2z)$. Find (i) a basis and dimension of Range(T), (ii) a basis and dimension of ker(T). Verify the dimension theorem. [MATH 215 | L-2/T-2 | 2018-19] **(30 marks)**

Q285. Let $S = \{v_1, v_2, v_3\}$ be a basis for $P_2(x)$ where $v_1 = 1 + 2x + x^2$, $v_2 = 2 + 9x$, $v_3 = 3 + 3x + 4x^2$. Find the coordinate vector of $p(x) = 6 + 6x + 3x^2$ relative to S. [MATH 247 | 2018–19]

Q286. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y, s, t) = (4x + y - 2s - 3t,\ 2x + y + s - 4t,\ 6x - 9s + 9t)$. Find a basis and the dimension of (i) the Kernel of T and (ii) the Range of T. [MATH 259 | 2018–19, 2022–23 (BUET)]

Q287. Determine whether the linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $w_1 = x_1 + 2x_2 + x_3$, $w_2 = -2x_1 + x_2 + 4x_3$, $w_3 = 7x_1 + 4x_2 - 5x_3$ is one-to-one. If so, find the inverse standard matrix and T^{-1}. [MATH 259 | L-2/T-1 | 2017-18] **(17 marks)**

Q288. If $T : V \to W$ is a linear transformation, prove that (i) the kernel of T is a subspace of V; (ii) the range of T is a subspace of W. [MATH 247 | L-2/T-2 | 2017-18] **(16.67 marks)**

Q289. Let $v_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $v_2 = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, and let $A = \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix}$ be the matrix for $T : \mathbb{R}^2 \to \mathbb{R}^2$ with respect to the basis $B = \{v_1, v_2\}$. (i) Find $[T(v_1)]_B$ and $[T(v_2)]_B$. (ii) Find $T(v_1)$ and $T(v_2)$. (iii) Find a formula for $T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. (iv) Compute $T\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. [MATH 247 | L-2/T-2 | 2017-18] **(20 marks)**

Q290. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by $T(w, x, y, z) = (2w + 4x + 6y + 5z,\ -w - 2x + 2y,\ 8y + 4z)$. Find a basis for (i) the kernel of T and (ii) the range of T. [MATH 215 | L-2/T-2 | 2017-18] **(17 marks)**

Q291. Let $B = \{(1, 3), (-2, -2)\}$ and $B' = \{(-12, 0), (-4, 4)\}$ be two bases for $\mathbb{R}^2$, and let $A = \begin{pmatrix} 3 & 2 \\ 0 & 4 \end{pmatrix}$ be the matrix for $T : \mathbb{R}^2 \to \mathbb{R}^2$ relative to B. (i) Find transition matrix P from B' to B. (ii) Find $[v]_B$ and $[T(v)]_B$ where $[v]_{B'} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. (iii) Find A' and P^{-1}. (iv) Find $[T(v)]_{B'}$. [MATH 215 | L-2/T-2 | 2017-18] **(20 marks)**

Q292. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x, y, z) = (x + 2y - z,\ 2x + y + z,\ y + z)$. Find the rank and nullity of T. [MATH 247 | L-2/T-2 | 2017-18] **(15 marks)**

Q293. Determine whether the mapping $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + 2y - z,\ y + z,\ x + y - 2z)$ is a linear operator. If so, find the basis and dimension of (i) $\operatorname{Im} T$ and (ii) $\ker T$. [MATH 247 | L-2/T-2 | 2017-18] **(15 marks)**

Q294. Find eigenvalues and corresponding eigenvectors of the linear transformation T on $\mathbb{R}^3$ defined by the reflection about the zx-plane. Is the transformation one-to-one? [MATH 259 | L-2/T-1 | 2017-18] **(10 marks)**

Q295. Find the standard matrix for the transformation T on $\mathbb{R}^3$, where T is the composition of a counterclockwise rotation of $\frac{\pi}{3}$ about the y-axis, followed by a reflection about the yz-plane, followed by contraction with factor $\frac{1}{2}$. Then find the image of $(2, 0, 3)$. [MATH 215 | L-2/T-2 | 2017-18] **(15 marks)**

Q296. Let $v_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $v_2 = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, and let $A = \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix}$ be the matrix for $T : \mathbb{R}^2 \to \mathbb{R}^2$ relative to basis $B = \{v_1, v_2\}$. (i) Find $[T(v_1)]_B$ and $[T(v_2)]_B$. (ii) Find $T(v_1)$ and $T(v_2)$. (iii) Find a formula for $T(x_1, x_2)^T$. (iv) Compute $T(1, 1)^T$. [MATH 247 | 2017–18]

Q297. Derive the standard matrices for the operations on $\mathbb{R}^3$: a rotation of $45°$ about the y-axis, followed by an orthogonal projection on the yz-plane, followed by a dilation with $k = 2$. Find the standard matrix for the composition and the image of $(-1, -2, 3)$. Is the operator one-to-one? [MATH 259 | L-2/T-1 | 2016-17] **(20 marks)**

Q298. Find the standard matrix for the linear operator $T : \mathbb{R}^3 \to \mathbb{R}^3$ given by $w_1 = 3x + 5y - z$, $w_2 = 4x - y + z$, $w_3 = 3x + 2y - z$. Then calculate $T(-1, 2, 4)$. [MATH 215 | L-2/T-2 | 2016-17] **(15 marks)**

Q299. Let $T : P_2 \to P_3$ be the linear transformation defined by $T(a + bx + cx^2) = x\big(a + b(x - 3) + c(x - 3)^2\big)$. Find the matrix for T with respect to $E = \{1, x, x^2\}$ and $E' = \{1, x, x^2, x^3\}$. Verify $A[p]_E = [T(p)]_{E'}$. [MATH 215 | L-2/T-2 | 2016-17] **(15 marks)**

Q300. Consider the basis $S = \{v_1, v_2, v_3\}$ for $\mathbb{R}^3$, where $v_1 = (1,1,1)$, $v_2 = (1,1,0)$, $v_3 = (1,0,0)$. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ with $T(v_1) = (1,0)$, $T(v_2) = (2,-1)$, $T(v_3) = (4,3)$. Find a formula for $T(x_1, x_2, x_3)$ and compute $T(5,-3,2)$. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**
↪ *Similar: MATH 259 | 2014-15, 2020-21, 2021-22; MATH 215 | 2016-17; MATH 143 | 2021-22, 2022-23*

Q301. Let T be multiplication by $A = \begin{pmatrix} 4 & 1 & 5 & 2 \\ 1 & 2 & 3 & 0 \end{pmatrix}$. Find a basis for Range(T), ker(T), the rank and nullity of T. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**
↪ *Similar: MATH 259 | 2014-15, 2018-19, 2021-22*

Q302. Let l be the line in the xy-plane through the origin making angle θ with the positive x-axis ($0 \le \theta < \pi$). Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ map each vector to its orthogonal projection on l. (i) Find the standard matrix for T. (ii) Find the projection of $X = (2,5)$ onto the line with $\theta = \pi/3$. [MATH 259 | L-2/T-1 | 2015-16] **(15 marks)**

Q303. Determine whether multiplication by $A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 3 & -4 \end{pmatrix}$ is one-to-one. Explain. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**

Q304. Find the eigenvalues and corresponding eigenvectors of $T : \mathbb{R}^2 \to \mathbb{R}^2$ where T is the reflection about the line $y = x$. Check your conclusion from the standard matrix. [MATH 259 | L-2/T-1 | 2015-16] **(10 marks)**

Q305. Find the standard matrix for the composition in $\mathbb{R}^3$: a rotation of 30° about the y-axis, followed by a reflection about the xy-plane, followed by an orthogonal projection on the yz-plane. Find the image of $(-1,2,3)$. [MATH 243 | L-2/T-2 | 2015-16] **(15 marks)**
↪ *Similar: MATH 247 | 2019-20; MATH 143 | 2022-23*

Q306. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x,y,z) = (x+2y,\ y-z,\ x+2z)$. Find the rank and nullity of T. Hence verify the Dimension Theorem. [MATH 247 | L-2/T-2 | 2015-16] **(15 marks)**

Q307. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation where $T(x,y,z) = (x+2y+z,\ y+z,\ x+y-2z)$. Find the basis of Im($T$) and ker($T$). Also find rank and nullity of T. [MATH 247 | L-2/T-2 | 2014-15] **(15 marks)**

Q308. Derive the standard matrices for the operations on $\mathbb{R}^3$: a rotation of $-30°$ about the x-axis, followed by a reflection about the yz-plane, followed by an orthogonal projection on the zx-plane. Find the standard matrix and the image of the triangle with vertices $(-1,2,-3)$, $(1,-2,-3)$, $(-1,-2,3)$. [MATH 259 | L-2/T-1 | 2013-14] **(20 marks)**
↪ *Similar: MATH 259 | 2014-15, 2016-17, 2017-18, 2018-19, 2021-22*

Q309. Determine the eigenvalues and corresponding eigenvectors of $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined as reflection about the yz-plane. Check your conclusion by computing from the standard matrix. [MATH 259 | L-2/T-1 | 2013-14] **(10 marks)**
↪ *Similar: MATH 259 | 2014-15, 2017-18*

Q310. Let $T : P_2 \to P_3$ be the linear transformation defined by $T(p(x)) = x \cdot p(x-3)$. Find the matrix for T with respect to the standard bases $B = \{1, x, x^2\}$ and $B' = \{1, x, x^2, x^3\}$. Verify the relation $A[p]_B = [T(p)]_{B'}$. [MATH 243 | L-2/T-2 | 2013-14] **(15 marks)**
↪ *Similar: MATH 215 | 2016-17*

Q311. (a) Determine whether the function $T : M_{22} \to \Re$ defined by $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = 3a - 4b + c - d$ is linear transformation or not. Justify your answer. **(10 marks)**
(b) If $T : V \to W$ is linear transformation, then prove that (i) The kernel of T is a subspace of V and (ii) The range of T is a subspace of W. **(15 marks)** [MATH 259 | L-2/T-1 | 2013-14]

Q312. Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $F(x,y,z) = (x+2y-3z,\ 2x+5y-4z,\ x+4y+z)$. Find a basis and dimension of the kernel and image of F. [MATH 243 | L-2/T-2 | 2012-13] **(15 marks)**

Q313. Consider the basis $S = \{u, v, w\}$ for $\mathbb{R}^3$, where $u = (1,1,1)$, $v = (1,1,0)$, $w = (1,0,0)$. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ with $T(u) = (2,-1,4)$, $T(v) = (3,0,1)$, $T(w) = (-1,5,1)$. Find a formula for $T(x,y,z)$ and compute $T(2,-3,5)$. [MATH 243 | L-2/T-2 | 2011-12] **(15 marks)**
↪ *Similar: MATH 243 | 2016-17*

BUET · Linear Algebra & Matrix Theory

Linear Algebra & Matrix Theory

Complex Vectors

Hermitian, Skew-Hermitian, Unitary Matrices, Spectral Theorem

S7 Complex Vectors

Q314. Define Hermitian and skew-Hermitian matrices. Examine whether the eigenvalues of the Hermitian matrix $H = \begin{pmatrix} 3 & 2-3i \\ 2+3i & -1 \end{pmatrix}$ are real. If so, verify that eigenvectors from different eigenspaces are orthogonal. [MATH 143 | L-1/T-2 | 2023-24] **(12 marks)**

Q315. Find the singular value decomposition (SVD) of $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$. [MATH 143 | L-1/T-2 | 2023-24, 2021–22] **(12 marks)**

↪ *Similar: MATH 143 | 2022-23*

Q316. Show that the Hermitian matrix $A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$ has real eigenvalues and that eigenvectors from different eigenspaces are orthogonal. [MATH 143 | L-1/T-2 | 2022-23] **(12 marks)**

Q317. Find a singular value decomposition (SVD) of the matrix $A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}$. [MATH 143 | L-1/T-2 |] **(17 marks)**

Q318. Find a matrix P that unitarily diagonalizes the Hermitian matrix $A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$. Verify that P^*AP is a diagonal matrix. [MATH 143 | L-1/T-2 | 2021-22] **(12 marks)**

Q319. Define Hermitian and Skew-Hermitian matrices with examples. Prove that any square matrix can be uniquely expressed as the sum of a symmetric and a skew-symmetric matrix. Verify for $A = \begin{pmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{pmatrix}$. [MATH 259 | L-2/T-1 | 2020-21] **(18 marks)**

Q320. Define orthogonal matrix and unitary matrix with examples. Express $A = \begin{pmatrix} -2+3i & 1-i & 2+i \\ 3 & 4-5i & 5 \\ 1 & 1+i & -2+2i \end{pmatrix}$ as a sum of Hermitian and skew-Hermitian matrices. [MATH 259 | L-2/T-1 | 2017-18] **(15 marks)**

Q321. Prove that every square matrix can be uniquely expressed as $P + iQ$, where P and Q are Hermitian matrices. [MATH 259 | L-2/T-1 | 2016-17] **(10 marks)**

↪ *Similar: MATH 259 | 2017-18, 2020-21*

Q322. Define Hermitian matrix and skew-Hermitian matrix with examples. Prove that every square matrix can be uniquely expressed as $P + iQ$, where P and Q are Hermitian matrices. [MATH 259 | L-2/T-1 | 2016-17] **(10 marks)**

↪ *Similar: MATH 259 | 2017-18, 2020-21*